



RESEARCH PROJECT I REPORT

**Photoluminescence Spectroscopy under high
magnetic fields**

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1 Introduction

This report describes the work done over the course of Research Project I (PH4112) at IISER Kolkata. The project was carried out under the supervision of Prof. Bhavtosh Bansal at the Pulsed Magnet Lab, IISER Kolkata.

1.1 Overview of Photoluminescence Spectroscopy

Photoluminescence spectroscopy is a non-destructive method used for probing the electronic structure of materials. It is based on the optical excitation of charge carriers across a bandgap in the material using lasers, which is followed by a relaxation and radiative recombination of the carrier and hole. The radiation emitted in this process is then analysed using standard spectroscopic techniques to obtain information about the band structure of the material being investigated.

The term “luminescence” in this context should be understood to be the surplus of electromagnetic radiation emitted by the material over its equilibrium radiation (which is predicted by Planck’s Law). This means that from a thermodynamic point of view luminescence is a non-equilibrium radiation. In addition to this, the radiation has to have a decay time that is much longer than the period of light oscillations ($10^{-14} - 10^{-15} \text{ s}$). [7].

Moreover, the term “photoluminescence” refers to the emission of radiation due to the excitation of charge carriers in a material by radiation of wavelength λ_{ex} , which is usually shorter than the emission wavelength λ_{em} [7].

Remark: Stokes Law

The relation,

$$\lambda_{ex} < \lambda_{em} \quad (1)$$

is called **Stokes Law** (or **Stokes Shift**). This captures the fact that some of the excitation energy is lost in non-radiative processes before the radiative recombination occurs.

The additional requirement for finite decay times for luminescent radiation is to differentiate it from other types of secondary radiation like reflected light, Cherenkov radiation, and various types of scattered light (Rayleigh, Raman, Brillouin scattering etc.). This distinction is important as the other types of secondary radiation mentioned originated due to very fast photon-matter interaction, and thus there is virtually no exchange of energy between the exciting photon and the electronic system of the material. On the other hand, during the process of

luminescence there is a genuine absorption of the excitation energy which causes a charge carrier to excite to a higher energy level - this necessitates a gradual change.

It should also be noted that the terms *phosphorescence* and *fluorescence* are historical terms that describe different ranges of decay times, and the term luminescence is a more general term that encompasses both of them.

The phenomena of luminescence can occur in most organic substances, and in many inorganic solids like ionic salts and semiconductors. The underlying reasons are different in organic and inorganic substances, and we will not be dealing with the organic case here.

1.1.1 Photoluminescence in Semiconductors

In semiconductors, the existence of photoluminescence is predicated by the existence of the forbidden energy gaps - which in turn only occurs if there is a cluster (several tens) of atoms present. Most bulk properties can be recovered in samples that are on the range of $1\mu m$ (or, 10^{10} atoms). Now, there are two kinds of luminescence in such materials,

1. Intrinsic Luminescence: Originates in ideal, pure, and defect-free lattices.
2. Extrinsic Luminescence: Originates in lattice defects and impurities.

The existence of the solid matrix in which the impurities exist affect extrinsic luminescence greatly - mostly by providing the statistical/mechanical dispersion of impurity centers and acting as an “antenna” for the extrinsic luminescence. We will not deal with the details of extrinsic luminescence here, and we will mostly regard our samples as pure and defect-free except where the analysis requires it.

Since we are concerned with spectroscopy of photoluminescence, the main goal is to analyse most of the characteristic features of the photoluminescence spectrum, and to assign a given spectral line to some process of radiative recombination.

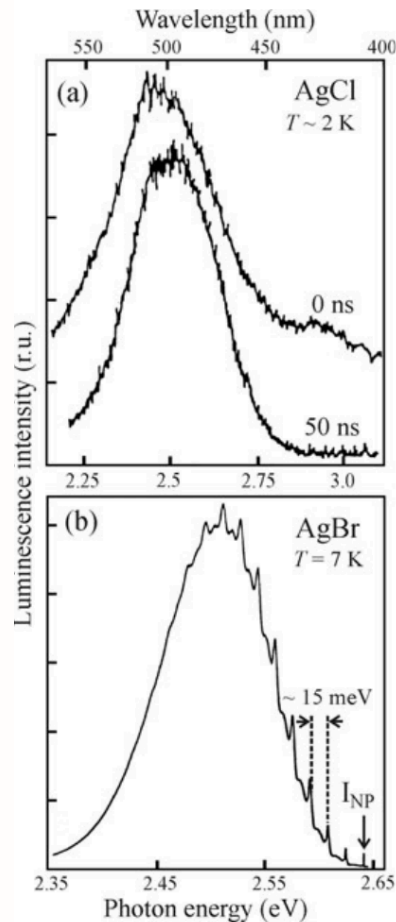


Figure 1: (a) PL spectra of pure $AgCl$ at liquid helium temperature under excitation by a pulsed laser ($\lambda_{ex} = 337\text{nm}$). The numbers near the curves denote delay after excitation. Source: [1] (b) PL spectra of nominally pure $AgBr$ at $T = 7\text{K}$. The fine structure superimposed on the band is due to interactions with phonons. Note that this band is entirely extrinsic in origin. Source: [2]

Remark: Silver Halides

While Silver halides are not typical semiconductors, they are used as a model system for PL spectroscopy as their optical and luminescence properties place them very close to polar wide-bandgap semiconductors.

Photoluminescence spectroscopy needs to be carried out at very low temperatures since most materials luminesce at low temperatures. Even if the material does luminesce at room temperature, it is often observed that the intensity of the luminescence increases with decrease in temperature. This contradicts the ordinary behaviour of thermal radiation in materials, and thus luminescence is sometimes known as **cold light**. The drop of luminescence intensity with increasing temperature is termed **thermal quenching**.

Finally, in semiconductor luminescence it is possible to have **edge emission**. This means that there is luminescent radiation that is situated in the spectral range

around the (optical absorption) band-edge of the material. This means that the energy of the luminescence photon is slightly lower than the band-gap energy. This is normally consisted of a number of narrow excitonic lines (intrinsic and extrinsic).

1.2 Overview of High Magnetic Field Physics in the context of PL Spectroscopy

Mainly, the Pulsed Magnet lab here at IISER Kolkata deals with magneto-photoluminescence spectroscopy of materials at very low temperatures using the home-built Pulsed Magnet. More on the experimental setup will be provided in the instrumentation section.

magneto-photoluminescence is used to probe the electronic band structure of materials, effective masses, g-factors, and the symmetries of various excitonic states. Fundamentally, the application of an external magnetic field causes two major changes in the system,

1. It breaks **time-reversal symmetry**.
2. It lowers the **spatial symmetry** of the system.

This leads to a number of interesting effects in the system, such as [8],

- The Zeeman Effect (Linear regime)
- The Diamagnetic Shift (Quadratic regime)
- Landau Quantization in the high-field regime
- The Magneto-Stark effect

These effects can be exploited to extract useful information regarding the material being studied, and will be discussed in the relevant theoretical sections.

1.3 Objectives of the project

The research project was mainly focused on:

1. Understanding photoluminescence spectroscopy
2. Understanding high magnetic field physics in the context of photoluminescence spectroscopy
3. Obtaining experimental data for magneto-photoluminescence spectroscopy of bulk $GaAs$ to gain familiarity with the Pulsed Magnet.
4. Understanding data collection and Instrumentation
5. Investigating possible avenues of research in the physics of excitons in Cu_2O under high magnetic fields.
6. (Planned future work) Setting up a transmission spectroscopy experiment.

2 Theoretical Background

2.1 Charged particles in Magnetic fields

We begin by stating electric and magnetic fields in terms of electromagnetic potentials:

$$\begin{aligned}\vec{E} &= -\nabla\varphi - \frac{\partial\vec{A}}{\partial t} \\ \vec{B} &= \nabla \times \vec{A}\end{aligned}\tag{2}$$

A valid Lagrangian for a particle of charge q and mass m in such an electromagnetic field is given by:

$$L = \frac{1}{2}m\dot{\vec{x}}^2 + q\dot{\vec{x}} \cdot \vec{A} - q\varphi\tag{3}$$

This will reproduce the Lorentz force law, as expected.

Let us now restrict ourselves to motion of the particle in a **constant** magnetic field.

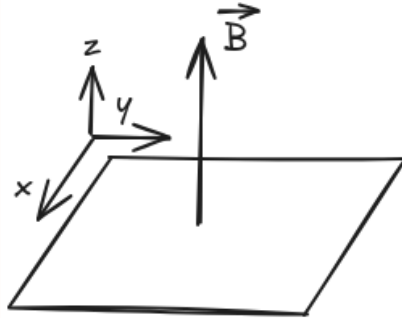


Figure 2: Let $\vec{B} = B\hat{z}$ and $\vec{E} = 0$. The charged particle is constrained to the XY plane shown. (Source: Author)

The equations of motion obtained are:

$$\begin{aligned}m\ddot{x} &= qB\dot{y} \\ m\ddot{y} &= -qB\dot{x}\end{aligned}\tag{4}$$

Which have the general solutions:

$$\begin{aligned}x(t) &= X + R\sin(\omega_B t) \\ y(t) &= Y + R\cos(\omega_B t)\end{aligned}\tag{5}$$

where X, Y, R are constants to be determined from the initial conditions, and $\omega_B = \frac{qB}{m}$ - the **Cyclotronic Frequency**.

Now, we may derive a canonical momentum from the prescribed Lagrangian (Equation 3):

$$\vec{\Pi} = \frac{\partial L}{\partial \dot{\vec{x}}} = m\dot{\vec{x}} + q\vec{A} \quad (6)$$

Note that this is *not* the same as the linear momentum $\vec{p} = m\dot{\vec{x}}$. Using this, we can perform a Legendre transform to arrive at a suitable Hamiltonian for the system:

$$\begin{aligned} H &= \dot{\vec{x}} \cdot \vec{\Pi} - L \\ &= \frac{1}{2m} (\vec{p} - q\vec{A})^2 + q\varphi \end{aligned} \quad (7)$$

Remark: Pierels substitution/Minimal coupling

The substitution $\vec{p} \rightarrow (\vec{p} - q\vec{A})$ is known as the **Pierels Substitution** or **Minimal coupling**. It usually refers to a coupling between fields which involves only the charge distribution and not the higher multipole moments of the charge distribution. This is in contrast to **Pauli coupling**, where the magnetic moment of the electron is also considered directly in the Lagrangian.

Note that the electromagnetic potentials have **gauge freedom**, i.e under the following transformations the electromagnetic field remains invariant:

$$\begin{aligned} \varphi &\rightarrow \varphi - \frac{\partial \alpha}{\partial t} \\ A &\rightarrow A + \nabla \alpha \end{aligned} \quad (8)$$

Where $\alpha(\vec{x}, t)$ is a scalar field. This means that there is a necessary redundancy in the description of the electromagnetic field. Introducing such a gauge field, we see that the linear momentum of the particle seems to undergo the transformation:

$$\vec{p} \rightarrow \vec{p} + q\nabla \alpha \quad (9)$$

We must conclude from this that the numerical value of $\vec{p}$ is not an observable quantity and has no physical significance. On the other hand, the canonical momentum $\vec{\Pi}$ and the hamiltonian H are observable quantities and have physical significance.

Promoting this discussion to a quantum treatment, we may write down the time-dependent Schrodinger equation (TDSE) for the system as:

$$i \hbar \frac{\partial \Psi}{\partial t} = H\Psi = \frac{1}{2m} (-i \hbar \nabla - q\vec{A})^2 \Psi + q\varphi \Psi \quad (10)$$

How does this equation fare under a gauge transformation? We note that to keep the TDSE invariant under the gauge transformation, the wavefunction must accumulate a phase under the transformation:

$$\Psi(\vec{x}, t) \rightarrow \exp\left(i \frac{q}{\hbar c} \alpha(\vec{x}, t)\right) \Psi(\vec{x}, t) \quad (11)$$

Since this is just a phase, no physical observables are affected by the gauge transformation - as is expected.

2.2 Landau Levels

Let us set $\varphi = 0$ and consider the Hamiltonian:

$$H = \frac{1}{2m} \left(-i \hbar \nabla - q \vec{A} \right)^2 \quad (12)$$

Now, we need to determine a gauge that we want to work in. Two solutions will be provided here.

2.2.1 Landau Gauge

The setup remains the same as shown in Figure 2. We choose a gauge where $\nabla \cdot \vec{A} = 0$ (the **Landau Gauge**). Hence, we have:

$$\vec{A} = xB\hat{y} \quad (13)$$

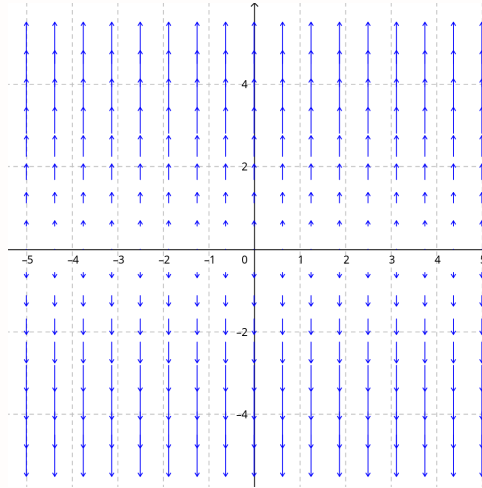


Figure 3: Plot of the vector potential $\vec{A}$ in the Landau Gauge. ($B = 5$) (Source: Author)

Clearly, in this gauge the field does not demonstrate translational invariance in the x-direction but it maintains the translational invariance in the y-direction. It also does not have rotational symmetry.

Now, we may expand the linear momentum operator in 3D to get:

$$H = \frac{1}{2m} \left[p_x^2 + (p_y - qBx)^2 + p_z^2 \right] \quad (14)$$

Note that $[p_y, H] = [p_z, H] = 0$ as there are no y or z operators in the Hamiltonian, which means that these are good quantum numbers. We thus make the ansatz:

$$\Psi(\vec{x}) = \exp(ik_y y + ik_z z) \chi(x) \quad (15)$$

Using this in the time-independent Schrodinger equation (TISE) assuming the energy eigenvalue to be E , we get:

$$\tilde{H} \chi(x) = \left(E - \frac{\hbar^2 k_z^2}{2m} \right) \chi(x) \quad (16)$$

Where $\tilde{H} = \frac{1}{2m} p_x^2 + \frac{1}{2m} (qBx - \hbar k_y)^2$. We can rearrange this Hamiltonian a bit to get it into a familiar form:

$$\tilde{H} = \frac{1}{2m} p_x^2 + \frac{1}{2} m \left(\frac{q^1 B^2}{m^2} \right) \left(x - \frac{\hbar k_y}{qB} \right)^2 \quad (17)$$

Setting $\omega_B^2 = \frac{q^2 B^2}{m^2}$ to be the cyclotron frequency, and $l_B = \sqrt{\frac{\hbar}{qB}}$ as the **magnetic length**, we have:

$$\tilde{H} = \frac{1}{2m} p_x^2 + \frac{1}{2} m \omega_B^2 (x - k_y l_B^2)^2 \quad (18)$$

This is simply the Hamiltonian for a Quantum Simple Harmonic Oscillator (QSHO) with a frequency ω_B and a mass m . The QSHO's "center" is determined by the eigenvalue k_y as can be seen.

Remark: Typical values of magnetic length l_B

For a magnetic field strength $B = 1T$ we have $l_B \approx 2.5 \times 10^{-8} m$ for an electron.

Thus, the energy eigenvalues of the full problem are:

$$E(n, k_z) = \hbar \omega_B \left(n + \frac{1}{2} \right) + \frac{(\hbar k_z)^2}{2m} \quad (19)$$

Where $n \in \mathbb{N}$ and $k_z \in \mathbb{R}$. Note that $k_y \in \mathbb{R}$ does not affect the energy eigenvalues. The wavefunctions are thus given by:

$$\Psi_{n, k_y, k_z}(x, y, z) \sim \exp(ik_y y + ik_z z) H_n(x - k_y l_B^2) \exp\left(-\frac{(x - k_y l_B^2)^2}{2l_B^2}\right) \quad (20)$$

Where H_n is the n^{th} Hermite polynomial. Stripping away the z dependence (which is immaterial anyways, seeing how the electron remains a free particle in that direction), we can plot the wavefunction to see that it is a Gaussian centered at $x = k_y l_B^2$ with a width of l_B . Since the wavefunction is a function of k_y while the energy eigenvalues are not, we can be certain that there is a **large degeneracy** of states here. Since the spectrum is identical to that of a QSHO otherwise, we can call all states labelled by the same value of n a **Landau Level** (LL). So, in the presence of the field, the continuous energy spectrum becomes discrete.

Remark: How large is this degeneracy of LLs?

Let us restrict ourselves to a finite region in the XY plane. Let this region be rectangular with the side length L_x, L_y . This causes a quantisation of the k_y eigenvalues (assuming suitable boundary conditions), giving us $k_y = \frac{2\pi}{L_y}m, m \in \mathbb{Z}$. But we can not quantise k_x as it is not a good quantum number. Since the wavefunction is localised around $x = k_y l_B^2$ and we have the condition $0 \leq x \leq L_x$, we have:

$$0 \leq k_y \leq \frac{L_x}{l_B^2} \quad (21)$$

as the allowed range for values of k_y . This means that the degeneracy is:

$$N = \frac{L_y}{2\pi} \int_0^{\frac{L_x}{l_B^2}} dk = \frac{L_x L_y}{2\pi l_B^2} = \frac{qBA}{2\pi \hbar} \quad (22)$$

Where A is the area of the region.

2.2.2 Symmetric Gauge

Note that the Landau gauge treats x and y coordinates assymmetrically, whereas there is no physical distinction between the two due to symmetry of the system. A more appropriate gauge turns out to be the choice such that:

$$\vec{A} = \frac{1}{2} \vec{B} \times \hat{r} = \frac{B}{2} [-y\hat{x} + x\hat{y}] \quad (23)$$

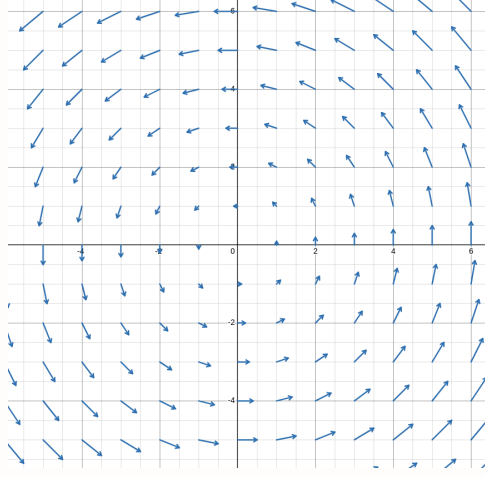


Figure 4: Plot of the vector potential $\vec{A}$ in the Symmetric Gauge. ($B = 2$) (Source: Author)

Now, translational symmetry is broken in the x- and y-directions, but we have full rotational symmetry about the z-axis. We can rearrange the Hamiltonian into the form:

$$H = \frac{1}{2m}p^2 + \frac{qB}{2m}L_z + \frac{q^2 B^2}{8m}(x^2 + y^2) \quad (24)$$

Where p is the total linear momentum operator, and L_z is the z-component of the angular momentum operator. We will skip the detailed solution of the problem here, but it may be found in [9]. In brief, restricting the electron to the XY plane have the energy eigenvalues:

$$E(n) = \hbar\omega_B \left(n + \frac{1}{2} \right) \quad (25)$$

Which is identical to the Landau gauge solution, and the wavefunctions are:

$$\Psi(x, y) = (x + iy)^m \exp\left(-\frac{x^2 + y^2}{4l_B^2}\right) \quad (26)$$

m represents the eigenvalues of the L_z operator.

This represents gaussian wavefunction with their peaks forming symmetric rings around the origin, with the ring being farther out for higher values of m . Limiting the rings using the physical size of the region we are working with, we can show that the expression for degeneracy of a LL is the exact same as Equation 22.

Remark: LLs and the Integer Hall Effect

Say we have a system with a fixed number of electrons. If we apply an external magnetic field, the electrons will fill up the LLs one by one, with the lowest energy LL being filled first. This process continues until all the electrons are filled. If the number of electrons is a multiple of the degeneracy of the LLs, the system will have its resistivity quantised. This is the Integer Hall Effect.

Problem 1: Equivalence of the wavefunctions in different gauges

An apparent problem here is the difference between the wavefunctions derived from the solution in symmetric and Landau gauges. Since it is necessary that the wavefunctions describe the same physical reality, how are they equivalent? It is possible that linear combinations of the wavefunctions from one gauge can be used to construct a wavefunction in the other gauge. This is a relevant question that was pointed out in discussions during the project, and will be investigated further.

2.2.3 Why is the study of Landau Levels relevant to magneto-PL spectroscopy?

In magneto-PL spectroscopy, we are interested in the energy levels of the excitons in a magnetic field. The excitons are the bound states of an electron and a hole, and they are affected by the magnetic field in the same way as the electrons. The energy levels of the excitons are thus given by the energy levels of the electrons, and the study of Landau Levels can help us understand the energy levels of the excitons in a magnetic field.

In high magnetic fields, the external magnetic potential competes with and eventually dominates the internal Coulomb potential of the exciton. This evolution transforms the PL spectrum from a probe of bound states (excitons) to a probe of free carrier kinetics (Landau quantization).

The transition between these regimes, particularly in the case of Rydberg excitons (to be discussed in the relevant section) provides a probe to observe **Quantum Chaos**, where the classical dynamics of the Hydrogen atom in a magnetic field becomes chaotic. Thus, Landau Levels form in an important regime (high-field) for studying many properties of excitons quantitatively.

2.3 de Haas - van Alphen Oscillations

This portion was eventually deemed irrelevant to the project objectives, but was initially studied to improve understanding of the effects of the external magnetic field on an electron gas. It is included (in brief) for completion, and more detailed analysis can be found in [3], and further information can be found in [10] and [11].

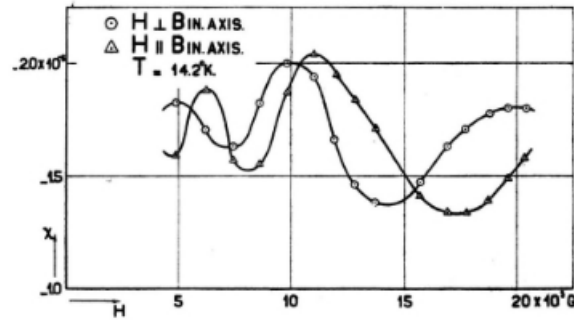


Figure 5: The first observation of the de Haas van Alphen effect, made by de Haas and van Alphen in 1930. Measures the magnetic susceptibility χ as a function of H . Source: The dependence of the susceptibility of diamagnetic metals upon the field, W. J. De Haas and P. M. Van Alphen, 1930

2.3.1 Landau diamagnetism

Landau diamagnetism is a purely quantum mechanical effect arising from the quantization of the orbital motion of conduction electrons in a magnetic field.

Theorem: Bohr-van Leeuwen Theorem

When statistical mechanics and classical mechanics are applied consistently, the thermal average of the magnetization is always zero.

This makes magnetism in solids solely a quantum mechanical effect and means that classical physics cannot account for paramagnetism, diamagnetism and ferromagnetism [12]. The inability of classical physics to explain triboelectricity also stems from the Bohr–Van Leeuwen theorem. Thus, it is a significant theorem and quite relevant to the topics being studied here

Proof: We will reproduce an *intuitive* and rough proof of the theorem here. A recent and more rigorous proof may be found here [13].

First, we state that the original theorem applies only to isolated systems that can not rotate. If an isolated system is allowed to rotate in response to the externally applied magnetic field, then the theorem does not apply.

The probability that the system will be in a given state of motion is predicted by Maxwell-Boltzmann statistics and is proportional to $\exp\left(-\frac{E}{k_B T}\right)$ where E is the energy of the system. Looking at the Lorentz force law, we can see:

$$\begin{aligned}\vec{F} &= q(\vec{E} + \vec{v} \times \vec{B}) \\ \Rightarrow \vec{F} \cdot \vec{v} &= q\vec{E} \cdot \vec{v}\end{aligned}\tag{27}$$

Which means that the rate of work done by the Lorentz force **does not depend on $\vec{B}$** . Therefore, the energy does not depend on the magnetic field, which implies that the distribution of motion in the system also does not depend on the magnetic field.

In zero field, there will be no net motion of the charged particles, as the system is not able to rotate. This will result in a net magnetic moment of zero. Since we have shown that the distribution of the motion does not depend on the magnetic field, the magnetic moment in thermal equilibrium will remain zero regardless of the external applied magnetic field.

□

Thus, diamagnetism only arises when we impose the discreteness of Landau Levels on the electron gas, which results in a modification to the density of states (DOS).

Let us start with an ideal Fermi gas of metal conduction band electrons. We ignore the spin dipole contributions, as that would lead to **Pauli diamagnetism**. The

magnetic field is setup the same as in Figure 2. In this case, we can begin with the usual Hamiltonian for a single electron:

$$H = \frac{1}{2m^*} (\vec{p} + e\vec{A})^2 \quad (28)$$

Where m^* is the effective mass of the electron in the metal. The energy eigenvalues are known to be quantized in the presence of a magnetic field, and are given by:

$$E(n, k_z) = \hbar \omega_B \left(n + \frac{1}{2} \right) + \frac{\hbar^2 k_z^2}{2m^*} \quad (29)$$

Here, $n \in \mathbb{N}$, $k_z \in \mathbb{R}$. The degeneracy per unit area perpendicular to the magnetic field is known to be $g = \frac{eB}{\hbar}$.

To find the magnetisation, we first calculate the Grand Canonical Potential Ω . For fermions, this is:

$$\Omega = -k_B T \sum_{\text{states}} \ln(1 + \exp(-\beta(E - \mu))) \quad (30)$$

Where $\beta = \frac{1}{k_B T}$, μ is the chemical potential, and E is the energy of the state. We now convert the sum over states to a sum over the LLs (i.e. n) and an integral over k_z . Set the dimensions of the Fermi gas to be (L_x, L_y, L_z) and corresponding degeneracy for each Landau Level to be $N = \frac{L_x L_y eB}{h}$. Putting these in:

$$\Omega = -k_B T \frac{L_x L_y eB}{h} \sum_{n=0}^{\infty} \int_{-\infty}^{\infty} \frac{L_z dk_z}{2\pi} \ln(1 + \exp(-\beta(E(n, k_z) - \mu))) \quad (31)$$

Let us now define $f(E)$ to be the function inside the integral, and $F(\hbar \omega_B (n + \frac{1}{2}))$ to be the integral of $f(E)$ carried out over k_z . We may express the Grand Canonical Potential as:

$$\Omega = \frac{eB}{h} \sum_{n=0}^{\infty} F\left(\hbar \omega_B \left(n + \frac{1}{2}\right)\right) \quad (32)$$

This is in the ideal form for the application of the Euler-Maclaurin Expansion.

Definition: Euler-Maclaurin Expansion Formula

$$\sum_{n=0}^{\infty} g\left(n + \frac{1}{2}\right) \approx \int_0^{\infty} g(x) dx + \frac{1}{24} g'(0) + \dots \quad (33)$$

Using the low field limit (i.e. $\hbar \omega_B \ll k_B T$), and considering the facts that the spacings between the LLs is $\Delta\varepsilon = \hbar \omega_B$, the integral term in the expansion corresponds to the zero-field free energy Ω_0 , and the first correction term carries the magnetic field contribution. Thus, the field-dependent correction to the potential is:

$$\Delta\Omega(B) \approx \frac{1}{24}(\hbar \omega_B)^2 \frac{\partial^2 \Omega_0}{\partial \mu^2} \quad (34)$$

Ω_0 is related to the DOS at the Fermi energy, so we have:

$$\Delta\Omega(B) = \frac{1}{6}\mu_B^2 B^2 D(E_F) \left(\frac{m_e}{m^*}\right)^2 \quad (35)$$

Approximately. Taking the appropriate derivatives of the potential, the expression for Landau susceptibility is found to be:

$$\chi_L = -\frac{1}{3}\mu_B^2 D(E_F) \left(\frac{m_e}{m^*}\right)^2 \quad (36)$$

Comparing to the expression for Pauli diamagnetism, we find that the Landau diamagnetism is $\frac{1}{3}$ of the Pauli diamagnetism. Exactly,

$$\chi_L = -\frac{1}{3}\chi_P \left(\frac{m_e}{m^*}\right)^2 \quad (37)$$

Where χ_P is the Pauli susceptibility. This essentially means that in a free electron gas, the Landau diamagnetic response cancels out $\frac{1}{3}$ of the paramagnetic response.

In semiconductors, if m^* is very small due to the band structure (seen in *GaAs* or *InSb*), then the Landau diamagnetic response will be larger than the paramagnetic response, effectively making the material diamagnetic.

2.3.2 Returning to dHvA oscillations

In the context of the Landau diamagnetic response, the dHvA oscillations are oscillatory corrections to the magnetisation that arises due to the sum over discrete Landau Levels not being perfectly approximated by the integral. While Landau Diamagnetism is the response of the total energy of the electron sea, the dHvA oscillations are the response of the electrons at the Fermi Surface of the material.

Without going into the details of the derivation (it can be found worked out in [14]), the oscillatory contribution to the grand canonical potential is given by:

$$\ln(Z_{osc}) \approx -\frac{3}{2}N \left(\frac{\alpha}{\varepsilon_F}\right)^{\frac{3}{2}} \sum_{l=1}^{\infty} \frac{(-1)^l}{l^{\frac{3}{2}}} \frac{\cos\left(\frac{l\pi\varepsilon_F}{\alpha} - \frac{\pi}{4}\right)}{\sinh\left(\frac{l\pi^2}{\alpha\beta}\right)} \quad (38)$$

Where $\alpha = \mu_B B$, and ε_F is the Fermi energy.

Remark

The derivation of this oscillatory term crucially depends on an initial simplification of an integral in the *Boltzmann Limit* (i.e. $k_B T \gg \varepsilon_F$).

From this, note that when $\alpha \approx k_B T$ (strong-field, low temperature), only the $l = 1$ term contributes significantly. We can thus derive the oscillatory contribution to the susceptibility as:

$$\chi_{osc} \approx \frac{3\pi}{2} \frac{N \mu_B^2}{\varepsilon_F} \frac{k_B T \varepsilon_F^{\frac{3}{2}}}{(\mu_B B)^{\frac{3}{2}}} \left(\frac{\sin\left(\frac{\pi \varepsilon_F}{\mu_B B} - \frac{\pi}{4}\right)}{\sinh\left(\frac{\pi^2 k_B T}{\mu_B B}\right)} \right) \quad (39)$$

which is oscillatory in $\frac{1}{B}$. Note that since the oscillation shows up in the GCP, it shows up in a number of different observables in the same fashion.

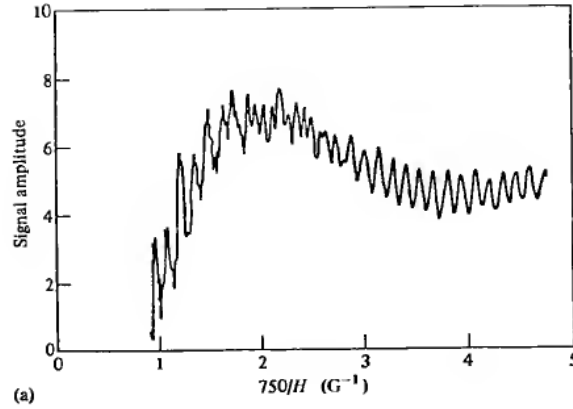


Figure 6: Oscillations in the sound attenuation in Tungsten, due to the same effects.

Source: C. K. Jones and J. A. Rayne, from [3]

2.3.3 Onsager's method for mapping the Fermi surface

Very briefly, we note very important use of the oscillations discussed above. Onsager [15] pointed out that these oscillations (alongside other types, like Shubnikov-de Haas oscillations) can be used to map the Fermi surface of a material. He put forward a simple formula to determine the area of the extremal cross-section of the Fermi surface perpendicular to the field direction (A_e) as:

$$\Delta\left(\frac{1}{B}\right) = \frac{2\pi e}{\hbar c} \frac{1}{A_e} \quad (40)$$

Where $\Delta\left(\frac{1}{B}\right)$ is the period of the oscillation.

2.4 The Hydrogen Atom

We go over, in brief, the relevant sections of the quantum mechanical solution to the Hydrogen atom problem as the objects of interest to us in this project (excitons) are hydrogen-like systems.

We begin with the simple Hamiltonian for the non-relativistic Hydrogen atom, which is the radial kinetic energy plus the Coulomb potential between the positively charged proton and the negative electron. We can put this into the TISE to get:

$$\left(-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \right) \psi(r, \theta, \varphi) = E\psi(r, \theta, \varphi) \quad (41)$$

Where $\mu = \frac{m_e M}{m_e + M}$ is the reduced mass of the electron-proton system, with M being the mass of the proton and m_e being the mass of the electron. In spherical polar coordinates, this is a separable partial differential equation. The solution can be found to be:

$$\psi_{nlm}(r, \theta, \varphi) = \sqrt{\left(\frac{2}{na_0^*} \right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-\frac{\rho}{2}} \rho^l L_{n-l-1}^{2l+1}(\rho) Y_l^{m(\theta, \varphi)} \quad (42)$$

Here, we have:

- $\rho = \frac{2r}{na_0^*}$
- $a_0^* = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2}$, which is the reduced Bohr radius.
- $L_{n-l-1}^{2l+1}(\rho)$ is a generalized Laguerre polynomial of degree $n-l-1$.
- $Y_l^{m(\theta, \varphi)}$ is the spherical harmonic function of degree l and order m .

Now, there are three good quantum numbers that label the state here (not consider the spin of the electron yet):

- n : The principal quantum number, which labels the energy level. $n = 1, 2, 3, \dots$
- l : The angular momentum quantum number, which labels the shape of the orbital.
 $l = 0, 1, 2, \dots, n-1$
- m : The magnetic quantum number, which labels the orientation of the orbital in space. $m = -l, \dots, 0, \dots, l$

In Dirac notation, we will be representing the state as $|n, l, m\rangle$. These states are properly normalized.

The analytical solution presented here relies heavily on the Coulomb potential being isotropic. Due to this, there is angular momentum conservation and rotational symmetry. This means that states with the same n but different l and m are degenerate. The energy eigenvalues are given by:

$$E_n = - \left(\frac{\mu e^2}{4\pi\epsilon_0 a_0^*} \right)^2 \left(\frac{1}{n^2} \right) = - \frac{Ry}{n^2} \approx - \frac{13.606 eV}{n^2} \quad (43)$$

Where Ry is the Rydberg. We will note that the amount of energy represented by a Rydberg changes in the case of other hydrogen-like system depending upon the values of the dielectric constant and effective mass is that system.

Of course, there is also the spin quantum number for the electron, s , which can take on the values $\frac{1}{2}$ or $-\frac{1}{2}$. This quantum number does not factor into the simple non-relativistic model of the Hydrogen atom that we have considered above, but it plays a crucial role in corrections to the model as we will discuss below. Considering this, the final degeneracy corresponding to the quantum number n is $g = 2n^2$.

2.4.1 Corrections to the Hydrogen Atom

In a real hydrogen atom (or hydrogen-like state, like excitons) there are several corrections that can be made. We will not be treating them extensively here, but we will list the important fine- and hyperfine-structure corrections, and justify them briefly.

2.4.1.1 The Fine Structure Corrections (H_{FS})

The Fine Structure correction is of the order $\alpha^2 E_n$, where $\alpha \approx 1/137$ is the fine structure constant. It arises from treating the electron relativistically. It consists of three distinct terms, treated below.

2.4.1.1.1 Relativistic Kinetic Energy Correction (H_{KE})

Justification: The Schrödinger equation uses $T = p^2/2m$, which is the non-relativistic dispersion relation. Special Relativity states the total energy is $E = \sqrt{p^2 c^2 + m^2 c^4}$. Expanding this for $p \ll mc$ (and removing the rest energy contribution mc^2):

$$\sqrt{p^2 c^2 + m^2 c^4} - mc^2 \approx \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} + \dots \quad (44)$$

The second term is the first-order correction.

Hamiltonian term added:

$$H_{KE} = - \frac{p^4}{8m^3 c^2} \quad (45)$$

Effect on Solution: Using first-order perturbation theory ($\langle H_{KE} \rangle = \langle \psi_n | H_{KE} | \psi_n \rangle$), this term lowers the energy of all states. The shift depends on both n and l .

2.4.1.1.2 Spin-Orbit Coupling (H_{SO})

Justification: In the rest frame of the electron, the proton orbits it, creating a magnetic field $\vec{B}$. The electron's intrinsic magnetic moment $\vec{\mu}_s$ interacts with this field (with the interaction being $-\vec{\mu}_s \cdot \vec{B}$). There is a relativistic kinematic correction factor of $\frac{1}{2}$ (Thomas Precession) because the electron's frame is non-inertial (rotating), and this nicely cancels out the Lande g-factor for the electron.

Hamiltonian term added:

$$H_{SO} = \frac{e^2}{8\pi\epsilon_0} \frac{1}{m^2 c^2 r^3} (\vec{L} \cdot \vec{S}) \quad (46)$$

Effect on Solution: Using the total angular momentum $\vec{J} = \vec{L} + \vec{S}$, and the identity $\vec{L} \cdot \vec{S} = \frac{1}{2}(\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$, this splits the energy levels for $l > 0$ based on $j = l \pm \frac{1}{2}$ (total angular momentum eigenvalues). States with higher j generally have higher energy.

Remark

This term is zero for s-states ($l = 0$) because angular momentum is zero.

2.4.1.1.3 The Darwin Term (H_D)

Justification: In Dirac theory, the electron is not a point charge but exhibits “Zitterbewegung” (rapid trembling motion) over a scale of the Compton wavelength $\lambda_c = \frac{\hbar}{mc}$. This is essentially due to the concept of point charges failing at lengthscales shorter than the lengthscale of the Compton wavelength (The energy uncertainty due to confinement at these lengthscales is enough to generate electron-positron pairs from vacuum). The electron effectively “feels” a smeared-out potential $\langle V(\vec{r} + \delta\vec{r}) \rangle$. This smearing only affects the energy where the potential curvature is high. This is specifically the case at the proton ($r = 0$).

Hamiltonian term added:

$$H_D = \frac{\hbar^2}{8m^2 c^2} \nabla^2 V = \frac{\hbar^2}{8m^2 c^2} \left(\frac{e^2}{\epsilon_0} \delta^{(3)}(\vec{r}) \right) \quad (47)$$

Effect on Solution: This term is non-zero **only for s-states** ($l = 0$), as only their wavefunctions are non-zero at the origin ($r = 0$). It pushes the energy of s-states up.

2.4.1.1.4 Total Fine Structure Result

Remarkably, when summing all three terms ($H_{KE} + H_{SO} + H_D$), the dependence on l cancels out partially, and the energy depends only on n and total angular momentum j :

$$E_{n,j} = E_n \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] \quad (48)$$

This formula lifts the degeneracy on j but leaves states like $2s_{\frac{1}{2}}$ ($l = 0, j = \frac{1}{2}$) and $2p_{\frac{1}{2}}$ ($l = 1, j = \frac{1}{2}$) degenerate.

2.4.1.1.5 The Lamb Shift (H_{Lamb})

Justification: This requires Quantum Electrodynamics (QED). It arises from the quantization of the electromagnetic field itself (and subsequent vacuum fluctuations). The electron constantly emits and re-absorbs virtual photons, which modifies its effective mass and vertex interaction. A detailed study of the reasons for this effect are beyond the scope of this project, for now.

Effect on Solution: It lifts the degeneracy between states with the same n and j but different l .

- Most notably, it splits the $2s_{\frac{1}{2}}$ and $2p_{\frac{1}{2}}$ levels.
- The $2s_{\frac{1}{2}}$ state is shifted upwards by ≈ 1.057 GHz (in frequency units) relative to $2p_{\frac{1}{2}}$.

2.4.1.2 Hyperfine Structure (H_{HF})

Justification: The proton is not just a point charge; it has a spin $\vec{I}$ and an associated magnetic moment $\vec{\mu}_I$. This nuclear magnetic moment interacts with the magnetic field generated by the electron (both orbital and spin contributions).

Hamiltonian term added (Fermi Contact Term + Dipole-Dipole): For s-states ($l = 0$), the interaction is purely the “contact” interaction (overlap of electron and proton):

$$H_{HF} = \frac{2\mu_0}{3} \vec{\mu}_e \cdot \vec{\mu}_I \delta^{(3)}(\vec{r}) \quad (49)$$

Since $\vec{\mu}_e = -g_e \mu_B \vec{S} / \hbar$ and $\vec{\mu}_I = g_I \mu_N \vec{I} / \hbar$:

$$H_{HF} \sim (\vec{I} \cdot \vec{S}) \quad (50)$$

Effect on Solution: We define the total atomic angular momentum $\vec{F} = \vec{J} + \vec{I}$. Using $\vec{I} \cdot \vec{S} = \frac{1}{2}(\vec{F}^2 - \vec{I}^2 - \vec{S}^2)$, the energy splits based on the alignment of nuclear and electron spins.

- For the Hydrogen ground state ($1s$), the proton ($I = \frac{1}{2}$) and electron ($S = \frac{1}{2}$) can form a triplet ($F = 1$, parallel spins) or a singlet ($F = 0$, anti-parallel spins).
- The transition between these two levels corresponds to the famous **21 cm line** (1420 MHz) used in radio astronomy.

2.4.1.3 Summary of Corrections

Correction	Order of Magnitude	Physical Origin	Dependency
Gross Structure	$E_n \sim \alpha^0$	Electrostatics	n
Fine Structure	$\sim \alpha^2 E_n$	Relativity and Spin	n, j
Lamb Shift	$\sim \alpha^3 E_n$	QED (Vacuum Fluctuations)	n, l, j
Hyperfine	$\sim \left(\frac{m_e}{m_p}\right) \alpha^2 E_n$	Nuclear Spin	n, l, j, F

Table 1: Hierarchy of Hydrogen Atom Energy Corrections

A detailed discussion regarding the solution of the hydrogen atom problem and all of the various correction can be found in [16] (Chapter 13).

2.5 Light-Matter interaction

A suitable semi-classical approach to the interaction of light with matter is to treat light as a classical electromagnetic wave (adequately represented by a time-dependent vector potential $\vec{A}$), and the electrons (or any other relevant charge carrier) as quantum mechanical particles. The Hamiltonian for such a system is the usual:

$$H = \frac{1}{2m} (\vec{p} - e\vec{A})^2 + V \quad (51)$$

Expanding the square and ignoring the term $\sim A^2$ as a small relativistic correction, we find that the perturbation is:

$$H' = \frac{e}{2m} \vec{p} \cdot \vec{A} \quad (52)$$

Definition: Fermi's Golden Rule

Fermi's Golden rule dictates the probability of transition from a state Ψ_m to another state Ψ_n due to a perturbation H' :

$$W_{m \rightarrow n} = \frac{2\pi}{\hbar} |\langle \Psi_n | H' | \Psi_m \rangle|^2 \delta(E_n - E_m - \hbar \omega) \quad (53)$$

Where $\hbar \omega$ represents the energy cost of transition from m state to n state due to the perturbation. Thus, it requires that the energy difference between the two states is equal to the energy of the photon absorbed by the electron, in our case.

An application of Fermi's Golden rule easily informs us of a few important conditions that need to be satisfied for the transition to be caused by incident EM radiation:

- $|E_n - E_m| = \hbar \omega$
- $\Delta m = 0, \pm 1$
- $\Delta l = \pm 1$

Where m and l are the usual hydrogen quantum numbers. These conditions are required for the matrix elements of the perturbation to be non-zero, and thus the transition probability dictated by Fermi's Golden Rule to be non-zero.

2.5.1 Light-matter interaction in semiconductors

In brief, inside a material like the semiconductor the discrete energy levels denoted by m, n in Equation 53 are converted into continuous energy bands. Therefore, to adjust the rule, we replace the delta function in Equation 53 with the DOS inside the material, and the matrix element is replaced by Block wavefunctions.

Note that to have an optically-induced transition of charge carriers between the bands of a semi-conductor, we need to conserve **both** energy and *crystal momentum*. This is simple to satisfy in direct bandgap semiconductors like GaAs, but in indirect band-gap semiconductors a phonon is required to help conserve the crystal momentum.

2.6 Excitons

An exciton is a bound state of an electron and an electron hole that are attracted to each other by the electrostatic Coulomb force. It is an electrically neutral quasiparticle that can transport energy without transporting net electric charge [17].

Simply put, when a photon of energy $\hbar \omega \geq E_g$ (E_g is the band gap) is absorbed by a semiconductor, an electron is excited from the valence band to the conduction band,

leaving behind a hole in the valence band. It is important to note that this “hole” is simply an unoccupied quantum mechanical electron state in the valence band, but with a *positive* charge (an analogue for positrons in semiconductors). The electron and hole are attracted to each other by the electrostatic Coulomb force, forming an exciton. This interaction between the excited electron and the hole lowers the total energy of the pair to slightly below the conduction band-edge, and thus forms a bound state within the band-gap.

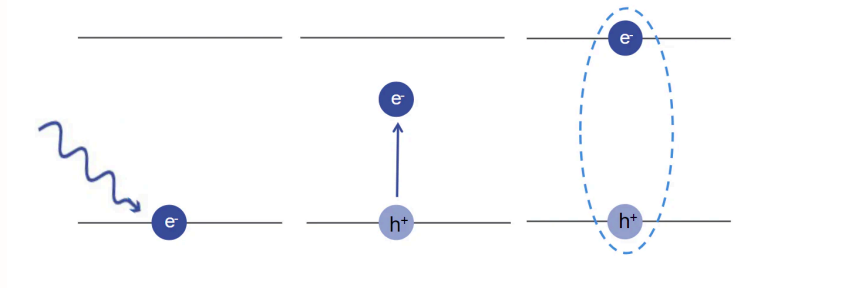


Figure 7: Diagram demonstrating the formation of an exciton in a semiconductor. Source: Osilla.com

Broadly, excitons are classified into two types based on the strength of the Coulomb interaction and the screening effects due to the underlying material:

2.6.1 Frenkel excitons

These are found in materials which has small dielectric constants (like organic crystals and alkali halides). Due to the small value of ϵ_r , the Coulomb interaction is strong and the binding energy is large (in the range of $\sim 0.1 - 1\text{eV}$). This leads to a small Bohr radius for the exciton, usually limiting them to within a unit cell. These can be thought of as localised quasiparticles which may hop from unit cell to unit cell. We will not be focusing on Frenkel excitons as these are not relevant (and do not show up) in the samples that we worked with.

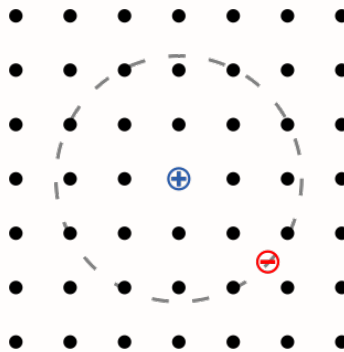


Figure 8: Diagram demonstrating a Frenkel exciton. Source: Wikipedia

2.6.2 Wannier-Mott excitons

In materials (such as most semiconductors) the dielectric constant is large. This increases the screening effect that is felt by the electrons and holes - resulting in weaker Coulomb interaction and smaller binding energies (on the order of 0.01eV). This type of exciton is called a Wannier-Mott exciton [18]. In most such material, the band structure also result is a lower effective mass of the electron (and hole). All these effects combine to result in an exciton that has a large Bohr radius which spans several unit cells.

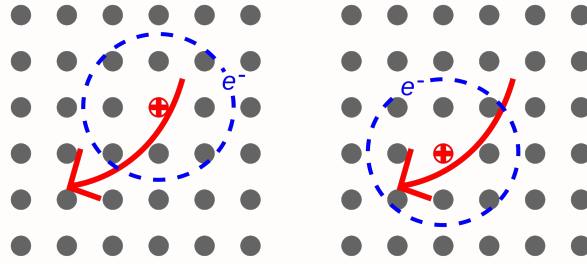


Figure 9: Diagram demonstrating a Wannier-Mott exciton. The diagram displays how the WM exciton is not localised and can diffuse through the lattice. Source: Wikipedia

Wannier-Mott (WM) excitons are considered as hydrogen-like bosonic (note that they are made out of two fermions) quasiparticles. Due to this, a lot of the physics of WM excitons is analogous to the physics of the Hydrogen atom problem discussed before.

We construct the Hamiltonian for the two-particle system (electron and hole) moving in the crystal lattice.

In the effective mass approximation, the periodic potential of the lattice is absorbed into the effective masses of the carriers (m_e^* and m_h^*). The Hamiltonian is:

$$H = -\frac{\hbar^2}{2m_e^*}\nabla_e^2 - \frac{\hbar^2}{2m_h^*}\nabla_h^2 - \frac{e^2}{4\pi\epsilon_0\epsilon_r|\vec{r}_e - \vec{r}_h|} \quad (54)$$

Where:

- $\vec{r}_e, \vec{r}_h$ are the position vectors of the electron and hole.
- ϵ_r is the static dielectric constant of the material.

To solve this, we transform coordinates from individual positions to the **Center of Mass (CM)** coordinate $\vec{R}$ and the **Relative** coordinate $\vec{r}$.

The coordinate transformations are:

$$\vec{r} = \vec{r}_e - \vec{r}_h \quad (55)$$

$$\vec{R} = \frac{m_e^* \vec{r}_e + m_h^* \vec{r}_h}{m_e^* + m_h^*} \quad (56)$$

The mass definitions are:

- **Total Mass (M):** $M = m_e^* + m_h^*$ (Governs CM motion).
- **Reduced Mass (μ):** $\frac{1}{\mu} = \frac{1}{m_e^*} + \frac{1}{m_h^*}$ (Governs relative motion).

Substituting these into the Hamiltonian allows us to separate H into two independent parts: $H = H_{CM} + H_{rel}$.

$$H = \left[-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 \right] + \left[-\frac{\hbar^2}{2\mu} \nabla_{\vec{r}}^2 - \frac{e^2}{4\pi\epsilon_0\epsilon_r r} \right] \quad (57)$$

The total wavefunction $\Psi(\vec{R}, \vec{r})$ is a product of the CM motion and relative motion:

$$\Psi(\vec{R}, \vec{r}) = \psi_{CM}(\vec{R}) \varphi_{rel}(\vec{r}) \quad (58)$$

1. Center of Mass Motion (Free Particle): The first term is simply the kinetic energy of a free particle with mass M .

$$\psi_{CM}(\vec{R}) = \frac{1}{\sqrt{V}} e^{i\vec{K} \cdot \vec{R}} \quad (59)$$

The energy eigenvalues are:

$$E_{CM}(K) = \frac{\hbar^2 K^2}{2M} \quad (60)$$

where $\vec{K}$ is the wavevector of the exciton (center of mass momentum). This propagation of the exciton through the crystal lattice must be in agreement with the **Bloch Theorem**.

2. Relative Motion (Hydrogen Atom) The second term is identical to the Hydrogen atom Hamiltonian, but with reduced mass μ and screened charge e^2/ϵ_r .

$$\left[-\frac{\hbar^2}{2\mu} \nabla_{\vec{r}}^2 - \frac{e^2}{4\pi\epsilon_0\epsilon_r r} \right] \varphi(\vec{r}) = E_{rel} \varphi(\vec{r}) \quad (61)$$

The eigenvalues for the relative motion are the quantized **Rydberg states** (effectively, hydrogen-like states with altered Rydberg constant):

$$E_n = -\frac{\mu e^4}{32\pi^2 (\epsilon_0 \epsilon_r)^2 \hbar^2} \frac{1}{n^2} = -\frac{R_y^*}{n^2} \quad (62)$$

where R_y^* is the **Exciton Rydberg energy** (Effective binding energy).

Total Energy of the Exciton: Combining the band gap energy E_g , the binding energy, and the kinetic energy:

$$E_{tot}(n, \vec{K}) = E_g - \frac{R_y^*}{n^2} + \frac{\hbar^2 K^2}{2M} \quad (63)$$

Exciton Bohr Radius (a_B^*): The characteristic size of the ground state exciton ($n = 1$) is:

$$a_B^* = a_H \frac{\epsilon_r}{\mu/m_0} \approx 0.53 \text{\AA} \cdot \epsilon_r \frac{m_0}{\mu} \quad (64)$$

Because ϵ_r is large and μ is usually small ($< m_0$), the exciton radius is much larger than the hydrogen atom.

2.6.3 Fine Structure of WM excitons: Spin and Exchange

The zeroth-order Hamiltonian for the exciton discussed above yields a Rydberg spectrum. However, in most materials (like the one we are mostly concerned with, Cu_2O), the observed energies deviate from this scaling. Phenomenologically, one may capture this deviation by introducing the Quantum defect parameter $\delta_{n,L}$ into the energy expression:

$$E(n, L) = E_g - \frac{R_y^*}{(n - \delta_{n,L})^2} \quad (65)$$

Where L is the angular momentum quantum number for the exciton. Precision spectroscopy has determined that the quantum defect is significant for s and p states of the exciton, but vanishes rapidly for high- L states ($d, f, \dots$) [19]. The physical origin of this defect lies in the breakdown of the parabolic band approximation, and the central-cell corrections where the dielectric screening is reduced at short lengthscales.

Remark: Hensel-Suzuki Hamiltonian

A detailed treatment of this problem requires a treatment using the **Hensel-Suzuki Hamiltonian**. The HS Hamiltonian is generally described as [20]:

$$H_{HS} = H_0 + H_{SO} + H(d) \quad (66)$$

Where H_0 is the zeroth order Hamiltonian discussed before.

The first additional term H_{SO} accounts for spin-orbit coupling between the hole spin ($\vec{S}_h$) and the **pseudo-spin** ($\vec{I}$). This leads to the coupled hole spin $\vec{j} = \vec{I} + \vec{s}_h$ with the quantum numbers $j = |I \pm s_h|$. It is given by:

$$H_{SO} = \frac{2}{3} \overline{\Delta} (1 + \vec{I} \cdot \vec{s}_h) \quad (67)$$

Where $\overline{\Delta}$ is the spin-orbit coupling strength.

The next term $H_d = H_d^s + H_d^c$ is the sum of the spherical term H_d^s and the cubic term H_d^c . We will not be discussing these correction terms in detail here, but they are discussed in [21] and [22] respectively. Essentially, these terms account for the non-spherically symmetric bands that exhibit warping and non-parabolicity.

Beyond the corrections discussed above, a very crucial correction to the excitonic spectra arises from the spin structure of the exciton constituents, and the **exchange interaction**.

The simple Hydrogen model neglects the spin of the electron and hole.

- **Electron Spin:** $s_e = \frac{1}{2}$.
- **Hole Spin:** This depends on the valence band. For a standard “heavy hole” band, the effective angular momentum is often treated as $j_h = \frac{3}{2}$. It may also be $\frac{1}{2}$ - depends on the band-structure.

Exchange Interaction (H_{exch}):

In the excitonic many-body picture, the electron in the conduction band and the hole in the valence band are both fermions. The Pauli Exclusion Principle (PEP) demands that the total wavefunction of the two-particle system be antisymmetric under particle exchange. Since the hole here is effectively an empty electron wavefunction with a positive charge, the interaction between the excited electron and the remaining valence electrons have to include an *exchange term*. This interaction critically depends on the spin configuration. The effective exchange Hamiltonian is:

$$H_{exc} = \Delta_{ex} \vec{S}_e \cdot \vec{S}_h \quad (68)$$

Where depending on the sign of the exchange coupling constant Δ_{ex} , the exchange interaction lifts the degeneracy between the singlet and the triplet states in this coupled spin basis.

This splits the exciton ground state into:

1. **Para-excitons (Singlet-like):** Spins anti-parallel. Usually optically “dark” or forbidden in dipole approximation. This represents $S = 0$ antisymmetric spin state.
2. **Ortho-excitons (Triplet-like):** Spins parallel. Usually optically active (“bright”). This represents $S = 1$ symmetric spin states.

In Cu_2O , the ground state is the Para-exciton, which is why the lowest energy transition is strictly forbidden and requires phonon assistance or quadrupole interactions. This is an important feature of Cu_2O excitons, and will be relevant to us later.

2.6.4 Exciton physics in magnetic fields

The application of an external magnetic field $\vec{B}$ is the most powerful tool for probing the excitonic fine structure. It introduces controlled symmetry breaking and energy shifts that reveal the internal quantum numbers of the states. The Hamiltonian in the presence of a magnetic field is:

$$H_B = H_0 + H_{\text{Zeeman}} + H_{\text{Dia}} + H_{\text{MS}} \quad (69)$$

2.6.4.1 The Linear Zeeman Effect

The linear interaction arises from the coupling of the magnetic field to the magnetic moments of the carriers. The Zeeman Hamiltonian is given by:

$$H_{\text{Zeeman}} = \mu_B \vec{B} \cdot (g_c \vec{S}_e + g_v \vec{S}_h + g_{\text{orb}} \vec{L}) \quad (70)$$

where μ_B is the Bohr magneton. For the 1S state, $L = 0$, so only spin terms contribute. Note that g_c, g_v, g_{orb} are the g-factors of the carrier, hole, and orbital angular momentum, respectively.

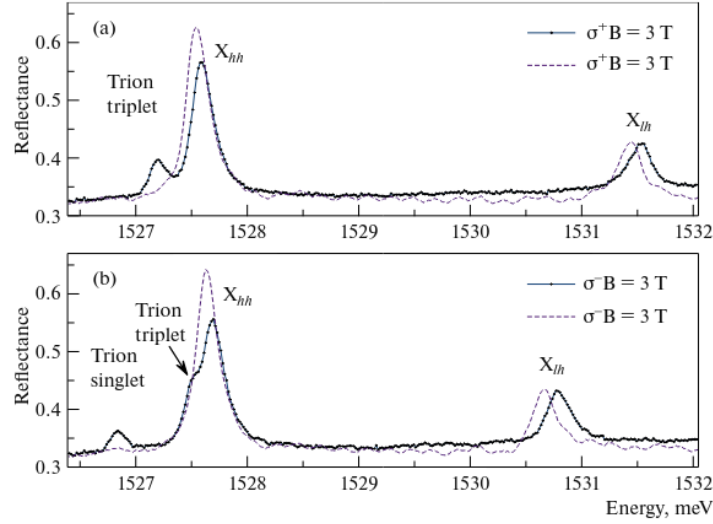


Figure 10: Reflection spectra of a heterostructure with a GaAs/AlGaAs quantum well of nominal width 14 nm: (a) σ^+ polarization; (b) σ^- polarization. Source: [4]

2.6.4.2 Splitting of the Ortho-Exciton

The triplet state (spin 1) splits into three components ($M_S = +1, 0, -1$) in a magnetic field. The energies shift as:

$$\Delta E = g_{\text{eff}} \mu_B B M_S \quad (71)$$

where $g_{\text{eff}} = g_c + g_v$. In Cu_2O , experimental values are $g_c \approx 2.02$ and $g_v \approx 3.5$ [19] (Zeeman components). The central component ($M_S = 0$) remains unshifted by the linear term.

2.6.4.3 Ortho-Para Mixing and Activation

A critical consequence of the Zeeman term is the mixing of the $M_S = 0$ ortho-state with the para-exciton singlet state ($S=0$). The eigenstates in a B-field are no longer pure singlets or triplets but linear combinations:

$$|\Psi\rangle = \alpha|S=0\rangle + \beta|S=1, M_S=0\rangle \quad (72)$$

The mixing coefficient depends on the difference in g-factors: $\Delta g = g_c - g_v$. The matrix element coupling the states is $\sim (g_c - g_v) \mu_B B$.

Because of this mixing, the “dark” para-exciton acquires a fraction of the oscillator strength from the “bright” (quadrupole-allowed) ortho-exciton. The absorption intensity of the para-exciton line scales as $I_{\text{para}} \sim B^2$. This magnetic-field-induced absorption appears approximately 12 meV below the ortho-exciton line, providing the most direct measurement of the exchange splitting energy.

2.6.4.4 The Diamagnetic Shift

The quadratic term in the vector potential $\vec{A} = \frac{1}{2}\vec{B} \times \vec{r}$ (in symmetric gauge) leads to the diamagnetic shift:

$$H_{\text{Dia}} = \frac{e^2}{8\mu} (\vec{B} \times \vec{r})^2 \quad (73)$$

This term always increases the energy of the state.

$$\Delta E_{\text{dia}} \sim \langle r^2 \rangle B^2 \quad (74)$$

For Rydberg excitons, the mean square radius scales as $\langle r^2 \rangle \sim n^4$. Consequently, the diamagnetic shift scales as $n^4 B^2$. Note this strong dependence - it becomes apparent why the high quantum number Rydberg excitons in Cu_2O are interesting.

For the ground state ($n = 1$), this shift is small ($\sim \mu\text{eV}$ at 10 T). However, for a giant Rydberg exciton with $n = 20$, the shift becomes massive, dominating the spectrum even at low fields. This shift pushes the high- n states toward the ionization continuum, eventually leading to the formation of Landau levels where the energy scales linearly with B (cyclotron motion dominance).

2.6.4.5 The Magneto-Stark Effect (Motional Stark Effect)

Perhaps the most fascinating phenomenon in the spectroscopy of giant Rydberg excitons is the Magneto-Stark effect (MSE). This effect arises from the relativity of electric and magnetic fields [23].

2.6.4.5.1 Mechanism

When an exciton with a finite center-of-mass momentum $\vec{K}$ moves through a magnetic field $\vec{B}$, it experiences an effective electric field $\vec{E}_{\text{eff}}$ in its rest frame:

$$\vec{E}_{\text{eff}} = \frac{\hbar \vec{K}}{M_{\text{exc}}} \times \vec{B} \quad (75)$$

where $M_{\text{exc}} = m_e + m_h$ is the total exciton mass. This electric field acts on the internal structure of the exciton, coupling to its electric dipole moment via the Stark Hamiltonian:

$$H_{\text{MS}} = e \vec{E}_{\text{eff}} \cdot \vec{r} = \frac{e \hbar}{M_{\text{exc}}} (\vec{K} \times \vec{B}) \cdot \vec{r} \quad (76)$$

2.6.4.6 Symmetry Breaking and State Mixing

Unlike the Zeeman effect (parity even) or Diamagnetic shift (parity even), the Stark operator ($\vec{r}$) has odd parity.

Parity Mixing: The MSE mixes states of opposite parity, such as S-states (even) and P-states (odd). In the absence of a field, P-states are dipole-allowed while S, D, F states are forbidden. The MSE enables the “dark” even-parity states (S, D) to borrow oscillator strength from the bright P-states.

Spectral Emergence: In a magnetic field, the absorption spectrum of high- n excitons transforms. Alongside the strong P-lines, new lines corresponding to S and D states emerge. Their intensity scales as $E_{\text{eff}}^2 \sim B^2 K^2$.

Scaling: The Stark effect scales with the polarizability of the state, which goes as n^7 .

$$\Delta E_{\text{MS}} \sim n^7 E_{\text{eff}}^2 \sim n^7 B^2 K^2 \quad (77)$$

This n^7 dependence makes the MSE the dominant perturbation for high Rydberg states. Even the tiny momentum of a photon $\vec{K}_{\text{photon}}$ is sufficient to generate an electric field that significantly perturbs states with $n > 20$.

There is a strong dependence of the geometry of the setup (relative orientation of $\vec{K}$ and $\vec{B}$) on the Stark effect. There two extreme cases are:

1. **Faraday Geometry** ($\vec{K} \parallel \vec{B}$): Here, the cross product $\vec{K} \times \vec{B} = 0$. The motional electric field vanishes, and the spectrum should show only the Zeeman splitting and diamagnetic shifts. Inversion symmetry is preserved.
2. **Voigt Geometry** ($\vec{K} \perp \vec{B}$): Here, the cross product $\vec{K} \times \vec{B}$ is maximum. The motional electric field is maximum, and the spectrum should show the Stark splitting. This allows for the observation of a “Stark Ladder” structure.

Currently at the Pulsed Magnet lab, we utilise a Faraday Geometry setup. A Voigt geometry setup is planned to probe the effects discussed.

2.7 Excitons in Cu_2O

To begin with, we must note the structure of the Cu_2O material. This material crystallizes in simple cubic lattice belonging to the space group $O_h^4(Pn\bar{3}m)$.

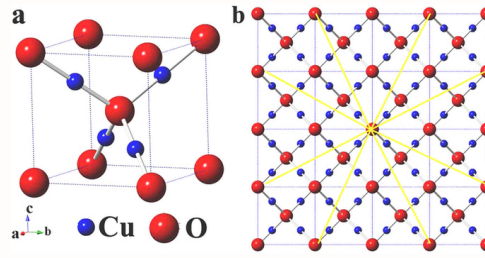


Figure 11: Crystal structure of Cu_2O . Source: ResearchGate, [5]

Definition: $O_h^4(Pn\bar{3}m)$ Space Group

The label $Pn\bar{3}m$, in Hermann-Mauguin notation (this can be found elaborated upon in [24], “Crystal Systems and Geometry”) is a recipe for the construction of the symmetry group. The equivalent Schonflies notation for this is O_h^4 . Term-by-term, this means:

- P (Primitive): The Bravais Lattice here is Simple Cubic. The primitive lattice vectors can thus be chosen to be Cartesian unit vectors that are scaled by the lattice constant a .
- n (Diagonal Glide Plane): It indicates a reflection plane (perpendicular to the principal axes), combined with a translation of half the face-diagonal.
- $\bar{3}$ (3-fold Roto-Inversion): The body diagonals of the cube (the $\langle 111 \rangle$ directions) are the axes of the 3-fold roto-inversion. - which is a rotation of 120° followed by inversion.
- m (Mirror Plane): There are two mirror planes perpendicular to the face diagonals ($\langle 110 \rangle$ directions).

To further describe the crystal mathematically, we attach a basis of atoms to the lattice points. For Cu_2O , the unit cell contains 2 Oxygen atoms and 4 Copper atoms.

We define the positions using fractional coordinates (u, v, w) such that the real positions are $\vec{r} = u\vec{a}_1 + v\vec{a}_2 + w\vec{a}_3$. ($\vec{a}_i$'s are the primitive lattice vectors).

- Oxygen sublattice: They form a BCC arrangement. The crystal does have a global inversion symmetry, but the Oxygen site locally does not. [19].
- Copper sublattice: They form a tetrahedron inside the cell (FCC sublattice, shifted by $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$), which is coordinated linearly between the Oxygens. The Copper site contains the center of inversion [19].

This space group information is important as it dictates the selection rules for excitations in the lattice (excitons, phonons, etc.), and the symmetry properties of any Bloch wavefunction on the lattice.

The “Yellow” series excitons that we are mainly concerned with (referring to [19]) originates from the fundamental gap between the Γ_6^+ conduction band and the Γ_7^+ valence band.

Remark: What do the band symbols Γ_6^+ and Γ_7^+ mean?

- Conduction band (Γ_6^+): This represents the lowest-energy conduction band at the center of the Brillouin zone, i.e. $k = 0$ or the Γ point. Originates from Copper 4s orbitals.
- Valence Band (Γ_7^+): This represents the highest-energy valence band at the Γ point. Originates from Copper 3d orbitals.

These symbols are derived from the irreducible representations of the cubic point group O_h - which is the symmetry group of the Cu_2O crystal lattice as discussed before. The indices of Γ label the specific transformation properties of the wavefunctions under the symmetry operators of the double group (including electron spin). Γ_6^+ corresponds to a spin-degenerate state ($S=1/2$) transforming like an $s_{\frac{1}{2}}$ atomic orbital. Γ_7^+ corresponds to a spin-degenerate state ($J = 1/2$) arising from the spin-orbit coupling of the d-orbitals.

Finally, the superscript $+$ represents the **parity**. In this case, it is even parity.

2.7.1 Orthoexcitons (Bright and Semi-Bright)

These are spin triplet states as discussed before. In Cu_2O , optical transitions must follow optical selection rules. Since the Γ_6^+ and Γ_7^+ bands are even parity, and the dipole operator is odd, a dipole transition is forbidden here in the atomic limit. However, p-type envelope function for $L = 1$ have odd parity, which compensates for the odd parity of the dipole operator, allowing for a dipole transition creating p-excitons.

Definition: Rydberg Series

This is a classic series of p-states that are observed upto $n = 30$ [19]. The linewidths scale as n^{-3} , and thus become very narrow for large n . This allows for a detailed resolution of the fine structure.

2.7.2 Paraexcitons (Dark States)

Paraexcitons are spin-singlet ($S=0$) states as discussed before. They are strictly optically forbidden in the electric dipole approximation due to spin-conservation and thus they are called *dark*.

They have to be primarily observed through two-photon absorption or by applying a magnetic field which causes a mixing between the ortho and para states.

2.7.3 Phenomena in magnetic fields

The excitons in Cu_2O exhibit all the phenomena that have been described previously in Section 2.6.4, but a particular phenomena not mentioned there that is of great interest to us is the **chaotic transition between the low-field and high-field (Landau) regimes**. As the principal quantum number n increases, the energy spacing between levels decreases as n^{-3} . Simultaneously, the magnetic interaction energy contributions from the Diamagnetic ($\sim n^4$) and Magneto-Stark ($\sim n^7$) effects also increase rapidly. There is a critical threshold, where the Coulomb binding energy and the magnetic interaction energies are of comparable magnitudes:

$$\gamma = \frac{\hbar \omega_B}{2R_y^*} \approx 1 \quad (78)$$

In this regime, the classical dynamics of the electron-hole pair is non-integrable and chaotic. This manifests in quantum mechanics as irregular spectral fluctuations. In Cu_2O this regime is accessible with moderate fields for $n > 5$. The clear hierarchy of the (n, l, m) quantum numbers collapse - the levels do not simply cross but exhibit level repulsion and anticrossings. Notably, the distribution of the energy level spacings transition in this regime from (uncorrelated) Poissonian distribution to Wigner-Dyson distribution. This systems serves as one of the rare easily realisable manifestations of quantum chaos. More study on this is planned for the future.

3 Instrumentation

Here, we document the various facets of the instrumentation available (and accessed for the project) at the Pulsed Magnet Lab. Much of the technical details have been extracted from [6], [25], and relevant device manuals from manufacturers.

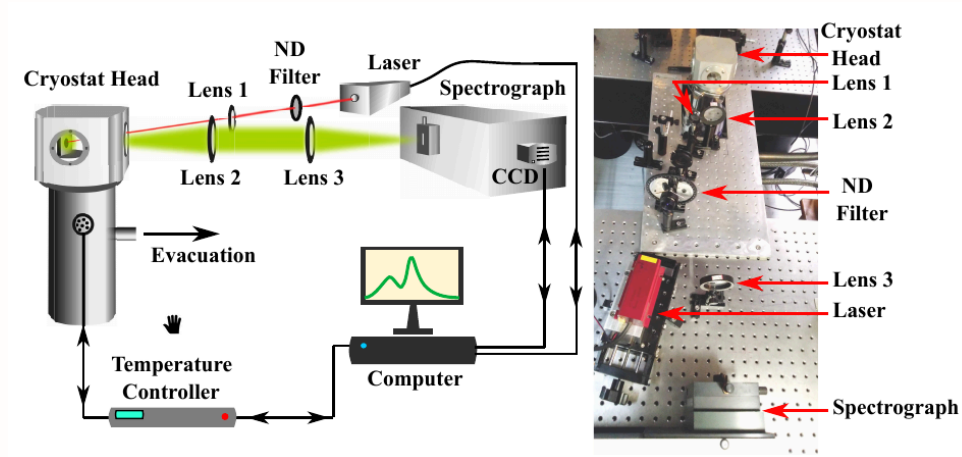


Figure 12: Schematic setup of PL spectroscopy in our Lab. Lenses and filters included in the diagram have not been used in context of this report. Source: [6]

3.1 Pulsed Magnet

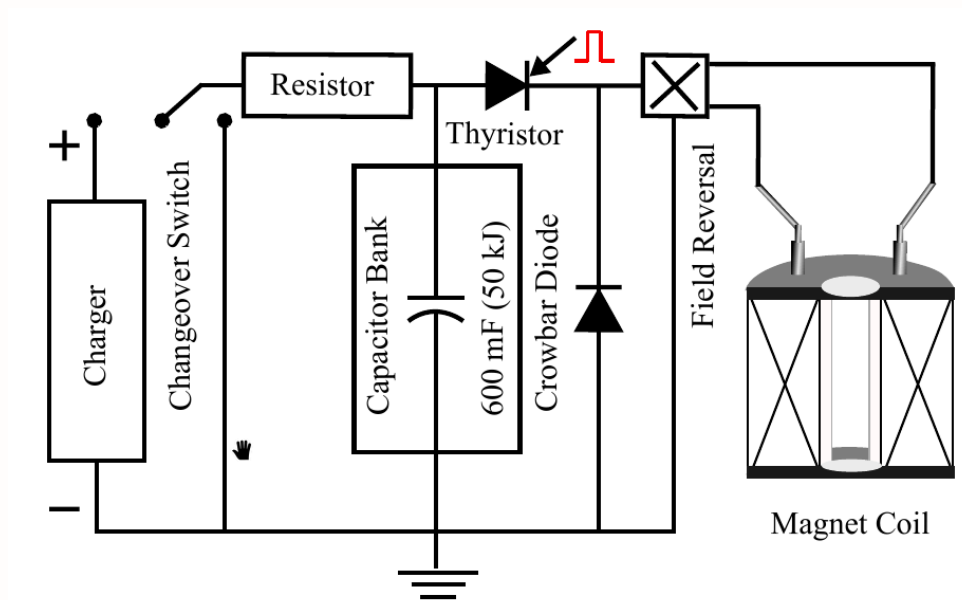


Figure 13: Schematic setup of Pulsed Magnet and associated discharge electronics. Source: [6]

Bore Diameter	18 mm
Length (z axis)	60 mm
Constructing Wire	Copper (8 Layers)
Wire Length	13.6 m
Reinforcement	Zylon (7 Layers)
Inductance	0.138 mH
Resistance (300 K)	23.71 mΩ
Resistance (77 K)	2.89 mΩ
Peak Field	35 T
Energy at Peak Field	48 kJ

Figure 14: Specifications of magnet coil. Source: [6]

The investigation of condensed matter systems under extreme environmental conditions serves as a fundamental probe into the quantum mechanical nature of solids. Among thermodynamic parameters such as temperature and pressure, the magnetic field occupies a unique position due to its ability to directly couple to the spin and orbital degrees of freedom of charge carriers. Magnetic fields act as a potent tool to lift spin degeneracies, quantize orbital motion into Landau levels, and compress exciton wavefunctions, thereby revealing parameters such as binding energies, reduced masses, and g-factors that are otherwise inaccessible.

While steady-state (DC) magnetic fields offer continuous measurement capabilities, their generation beyond 20-25 Tesla requires monumental infrastructure involving resistive-superconducting hybrid magnets, often consuming megawatts of electrical power and necessitating massive cooling plants. In contrast, pulsed magnetic field technology provides a viable, energy-efficient pathway to access ultra-high magnetic fields (30–100 T) in a compact laboratory setting [26].

The Pulsed Magnet at IISER Kolkata has a maximum field strength of 35 Teslas, and is driven by a 75 kJ Capacitor bank.

The construction of such a high-field pulsed magnet is an optimization problem balancing three competing effects/goals:

- Maximizing field for a given current
- Containing the immense Lorentz forces generated in the current carrying components
- Managing the heat load on the magnet

A multi-layer solenoidal coil is used to generate the field. If a_1 is the inner radius, and a_2 is the outer radius of the coil, we can calculate the field at the center of the coil as:

$$B_0 = \mu_0 \lambda j a_1 F(\alpha, \beta) \quad (79)$$

Where $\alpha = \frac{a_2}{a_1}$, β is the half-length of the solenoid, λ is the filling factor (ratio of conductor volume to the total coil volume), and $F(\alpha, \beta)$ is the **Fabry factor**, which

is a dimensionless deometry-dependent term and determines the efficiency of the coil. j is the current density in the coil.

The self-inductance L of the coil is the critical parameter that governs the temporal profile of the magnetic pulse that it generates. In a discharge circuit where the capacitor C is charged to a voltage V and discharged through the coil, the rise time t_{rise} (which is, the quarter of a period of oscillation in LC circuit), is determined by:

$$t_{rise} \approx \frac{\pi}{2} \sqrt{LC} \quad (80)$$

In the pulsed magnet, a specific coil geometry with a bore of 16 - 18 mm [25] is to generate a pulse which is approximately on the order of tens of ms. This choice of duration is long enough to minimize excessive eddy currents (Power dissipated by such eddy currents is $\sim \left(\frac{dB}{dt}\right)^2$) in the other metallic portions of the setup (such as the cryostat), but is short enough to minimise the resistive heating in the coil wire. In addition, the coil's inductance must be impedance matched to the capacitor bank to ensure maximum (and safe; mismatch causes large voltage spikes) energy transfer.

The Lorentz force density generated on the coil due to field is given by:

$$\vec{f} = \vec{j} \times \vec{B} \quad (81)$$

In a cylindrical solenoid, this manifests as primarily a radial stress that tends to push the coil outwards ($\sigma_{\theta\theta}$), and an axial compressive stress tending to crush the coil towards the midplane (σ_{zz}). The magnetic pressure of the field may be expressed as:

$$P_{mag} = \frac{B^2}{2\mu_0} \quad (82)$$

At 35 Teslas, the magnetic pressure is approximately **487 MPa**. The yield strength of annealed copper (standard) is only about 60-70 MPa, and for hardened copper it is 200-300 MPa. Thus, a pure copper coil would undergo plastic deformation at 35 Teslas. To withstand these pressures, the coil is reinforced using other materials. In addition, the reinforcement is not homogenous as the inner regions of the coil (where the field is higher) experience more stress.

The timescale of the pulse is significantly shorter than the thermal diffusion time constant of the coil's insulation material. Therefore, the heating of the coil during the pulse is to be treated as an adiabatic process, i.e. no heat is lost to the external liquid nitrogen bath during the pulse. The rise of temperature in the coil may be calculate/ approximated by accounting for the temperature dependent resistivity of the coil material, and the effects of magnetoresistance. Such a calculation would inform us of maximum safe current that the coil can carry - beyond which the heat can cause damage to the coil. We wil not attempt to carry out this calculation here.

3.1.1 Energy storage and discharge System

The pulsed magnet is driven using the capacitor bank which allows the discharge of a large amount of current/energy over a very small interval of time. The lab has a 75 kJ capacitor bank which can be charged using a Variac transformer and rectifier setup. The voltage of the bank at any point may be measured using a dedicated digital display. Note that the latency of the display and the manual nature of the charging apparatus often causes a variation of ± 10 V around the target voltage (which an overshoot being more common). The energy of the capacitor bank at some voltage can be calculated using the known formula:

$$E = \frac{1}{2}CV^2 \quad (83)$$

The bank is modular, and consists of multiple high-voltage rated capacitors that are connected in parallel via heavy-duty copper busbars to minimise the stray inductance and resistance.

Remark: Safety: Capacitor Bank

Since the capacitor bank can carry very high voltages, standard safety protocol in the lab prohibits anyone from being present inside the room housing the magnet unless the bank is grounded.

The discharge circuit forms an RLC loop comprising of the Capacitor Bank (C), the Magnet Coil (L, R), a high current switch, and a device to reverse the current flow.

- The switch is a high-current rated thyristor with an attached heat sink. It is triggered using a standard 5V TTL pulse from a microcontroller (currently, an Arduino Uno).

Another important part of the circuit is a diode stack placed in reverse bias over the coil - this acts as a **crowbar diode**. This provides the back-EMF from the collapsing magnetic field of the coil after a pulse peak a very short-resistance path and thus protects the capacitor bank.

3.2 Cryostat and Variable Temperature Insert (VTI)

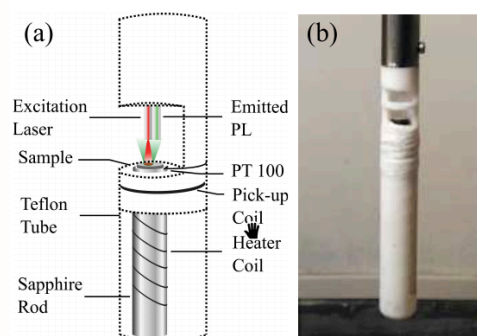


Figure 15: (a) Schematic of sample holder, (b) Image of sample holder. Source: [6]

While the magnet is held in a liquid nitrogen bath, the sample must be cooled to very low temperatures for useful spectral signatures to be detected. Moreover, a control over the sample temperature will also be useful to investigate a number of temperature dependent phenomena. This decoupling is done through a variable temperature insert (VTI).

The VTI is a double-walled vacuum-insulated tube that is inserted into the magnet bore. The outer diameter of the VTI is limited by the magnet bore (16-18 mm). The vacuum is maintained by a Wilson seal, and is created using a standard vacuum pump attached to a valve on the side.

The sample is mounted on top of a **sapphire cold finger**. Sapphire is the material of choice as it is an electrical insulator (prevents eddy currents) but has very high thermal conductivity at cryogenic temperatures. This ensures a good thermal equilibrium between the heater/sensor, cryogenic cooler, and sample. A heater coil made of a resistive material is wound around the cold-finger in a non-inductive manner. A calibrated PT100 Platinum temperature sensitive resistor is placed on the cold-finger in addition to this. A standard temperature controller measures the resistance of the PT100 using a 4-probe measurement setup, and controls the current through the heater coil to maintain a desired temperature.

A closed-cycle cryogenic cooler utilising Helium cools the cold-finger. The cooler functions by using the **Joule-Thomson Effect**.

3.3 Optical instrumentation: Magneto-PL Probe

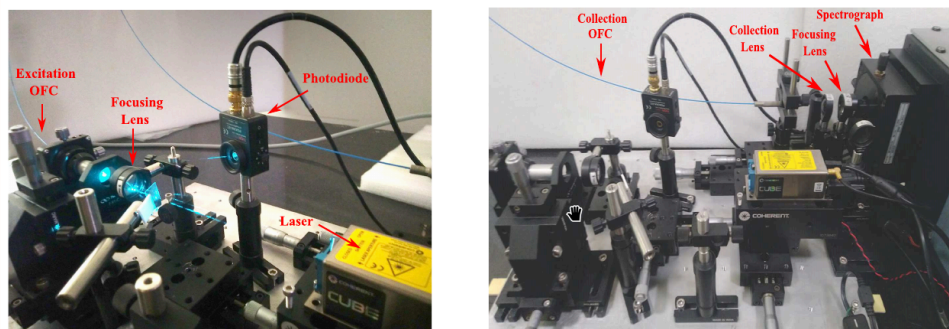


Figure 16: Picture of excitation optics (left) and collection optics (right). Source: [6]

Coupling light to-and-from the sample chamber inside the VTI is challenging due to the restricted access, and the mechanical vibrations associated with the pulse. To address this, multi-mode fiber optic cables are put through the Wilson seal on top of the VTI in a Faraday Geometry. This is an easily adjustable and vibration-misalignment free solution. There is one fiber that carries the excitation from the laser to the sample, and a collection fiber that is used to collect photoluminescence emitted from the sample and carry it to the Andor spectrometer.

The fiber optics terminate very close to the sample to maximize efficiency. There are no lenses or other optical elements present inside the chamber.

Remark: Polarisation-resolved spectroscopy

It is planned to introduce quarter-wave plates and polarisers inside the chamber between the sample and the fiber optics to enable us to carry out polarisation-resolved spectroscopy. Since particular exciton states couple to only specific polarisations of light, this will allow us to access more information about the exciton spectrum.

3.4 Photoluminescence Spectrometer

The spectral analysis is performed by the Andor Shamrock SR-303i, a compact Czerny-Turner spectrograph optimized for low-light applications.

3.4.1 Optical Configuration and Grating Tunnel

The SR-303i features a 303 mm focal length with an f/4 aperture, which is well-matched to the f-number of standard fiber optics, ensuring that the spectrometer is filled with light without excessive loss or stray light. A central feature is the Triple Grating Turret, which is motorized and interchangeable. This allows the user to switch between gratings with different groove densities without breaking vacuum or

realignment. The different gratings offer different blazings, and are thus optimised for different wavelength windows.

The entrance slit controls the spectral resolution and throughput. The SR-303i features motorized slits (Part no. SR-ASZ-0103) adjustable from 10 μm to 2.5 mm in 2.5 μm steps.¹ In magneto-PL, there is a constant trade-off: narrowing the slit improves resolution (necessary, for example to resolve sub-meV Zeeman splitting) but reduces signal intensity (problematic for short integration times). The motorized control allows for remote optimization of this trade-off during the experiment

3.4.1.1 The Detector

The facility uses an Andor iXon Electron Multiplying CCD (EMCCD), which is cooled using a Peltier module to reduce noise. Standard CCDs are limited by read noise at high readout speeds. The EMCCD overcomes this via an on-chip gain register. Electrons are shifted through a high-voltage register where impact ionization occurs, multiplying the signal electrons before they reach the readout amplifier. This effectively eliminates read noise, allowing for single-photon sensitivity even at the high frame rates required for pulsed fields.

An important mode of operations of the EMCCD camera is the **Fast Kinetics** mode. To capture several snapshots of the spectrum during a pulse (which significantly improves the throughput of data collection during experiments), the camera has to capture several spectra in a very small time interval. To achieve this, the camera exposes a very small region (user-defined, but typically set to 20 rows of pixels) to incoming light. Once the CCD is exposed and charge is accumulated, the 20 rows of charges are shifted *on the CCD chip* to lower rows. This frees up the top 20 rows immediately, and does not require us to wait for the long readout time. The CCD chip itself acts as an intermediate buffer region. This process of shifting into the buffer takes only about 1ms, and is thus good enough for our purposes. All the data is readout once the pulse is over.

3.5 Data Acquisition System

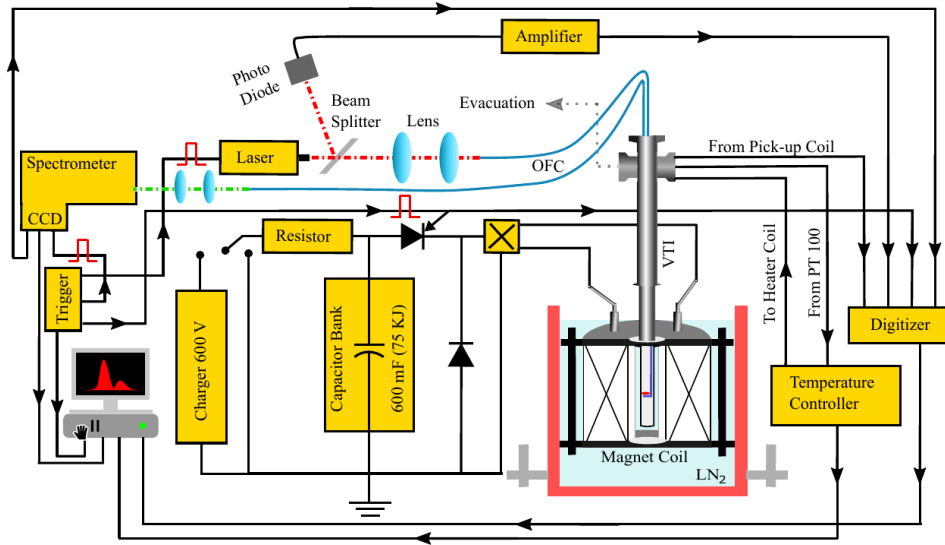


Figure 17: Schematic of entire magneto-PL setup with data acquisition devices.

Source: [6]

A pulsed field experiment is carried out in single-shot events. The discharge of the capacitor bank, the laser excitation, and the detector exposure must all be carefully coordinated and synchronized.

The facility current uses a standard Arduino Uno microcontroller with a 20 kHz clock as the master timing unit. The Arduino runs a routine (triggered via Arduino IDE on the computer) that coordinates the timing of the events in the experiment. The order of TTL trigger pulses sent out by the Arduino is as follows:

- T_0 : Trigger pulse sent to discharge thyristor.
- Now, several pulses (the number and delay between these pulses are user-defined) are sent to both the laser electronics, and then to the detector electronics after a slight delay. This ensures that the detector is exposed when the material is emitting photoluminescence.

The TTL (Transistor-Transistor Logic) level of the Arduino is 5V.

3.5.1 Pickup coil and Digitizer

Simultaneously with the optical data collection by the spectrograph, the magnetic field profile must be recorded. A calibrated pickup coil is wound around the sample position. By Faraday's Law,

$$V(t) \sim -NA \frac{dB}{dt} \quad (84)$$

where N is the number of turns in the pickup coil, A is the area of the coil. $A = 465.32 \times 10^{-6} m^2$ in our setup.

The coil is connected via BNC cables to an Agilent L4534A Digitiser, which acts as a very high throughput oscilloscope. The digitiser samples the pickup coil voltage at a very high (user-defined) frequency throughout the pulse and stores it on its hardware. The data can then be extracted to a computer via LAN, and integrated numerically to determine the magnetic field's temporal profile. The digitiser also has dedicated channels to record the trigger pulses sent out by the microcontroller, and this data may also be accessed over LAN.

3.5.2 Python-based DAQ and GUI for Digitizer

In the course of this project, some amount of time was dedicated to the creation of software that allows us to access data from the Digitiser, arm it, and carry out other functions over LAN. The proprietary software interface that Agilent provides is outdated, un-maintained, and requires Java plugins that most modern browsers have phased out. To address this, a software was created that could emulate the functions of the Agilent software interface. We will not delve into the technical details of the software and communication protocol used, but the code can be found on a publicly hosted Github Repository here: [Github Repository](#).

The code contains further technical documentation and instructions for use.

4 Experiment: Magneto-PL Spectroscopy of bulk $GaAs$

In this experiment, I aided senior PhD student Deepesh Kalauni in carrying out magneto-PL spectroscopy of a bulk $GaAs$ sample.

The optical properties of direct-bandgap semiconductors, particularly III-V compounds like Gallium Arsenide ($GaAs$), are governed at low temperatures by the formation and recombination of excitons. $GaAs$ has long served as the archetypal model system for the study of Wannier-Mott excitons due to its high crystalline purity, moderate direct bandgap ($\approx 1.519eV$ at liquid He temperatures [27]), and the isotropic nature of its conduction band minimum at the Γ -point (this reduces the effect of HS Hamiltonian fine-structure corrections). While zero-field PL measurement provide information about the band-gap and the impurity of the sample, the application of a field allows the extraction of intrinsic exciton parameters.

Important values relating to the material, extracted from literature [28]:

- Effective mass of the electron: $m_e^* = 0.067m_0$
- Effective mass of the hole (heavy hole): $m_h^* = 0.51m_0$
- Dielectric constant of $GaAs$ (static, not high-frequency): $\epsilon_r \approx 12.9$
- Lattice constant: $5.65325\text{\AA}$

Here, m_0 is the free electron mass. Using the following expression, we can calculate the effective Rydberg constant for the material:

$$R^* = R_0 \left(\frac{m_0}{m_e^*} \right) \left(\frac{1}{\epsilon_r^2} \right) \quad (85)$$

There $R_0 \approx 13.6eV$. The effective Rydberg constant for $GaAs$ comes out to be $\approx 4.2meV$. This is a great reduction in the Coulomb binding energy for the excitons. Also, the reduced mass of the exciton is $\approx 0.06m_0$. From literature values, the Bohr radius of the excitons (in zero-field) is in the range of 11 - 14 nm. From the energy scales noted here, it is easy to observe that cryogenic temperatures are crucial to observing the PL spectra of excitons in $GaAs$, as thermal energy at room temperature dwarfs the Coulomb energy of the excitons.

4.1 Experimental Procedure

The experiment was carried out in the Pulsed Magnet and VTI/Csryostat setup.

The sample was cleaned, prepared and mounted inside the Cryostat, and the cryostat seal was shut. The Magnet housing was put in a liquid nitrogen bath. Once this was done, the following procedure was followed to set up the experiment:

1. Connect the vacuum pump through silver tubing to the valve gate on the side of the cryostat. Release the knob to equalise pressure - note that the knob should be opened before the vacuum is pulled to avoid damage to the pump. If the cryostat is already under vacuum, the knob should be opened to equalise pressure before the pump is connected.
2. Wait for the pump barometer readout to reach at least 10^{-5} mbar.
3. Turn on the chiller for the cryogenic cooler in the adjoint room. This chiller circulates cold water to carry away heat from the cooler's radiator.
4. Wait for 5-10 minutes for the chiller to reach a stable temperature.
5. Turn on the compressor for the cryogenic cooler.
6. Observe the temperature readout on the connected temperature controller. Wait for the temperature to bottom out - with a nitrogen bath this is typically around $5K$.
7. Turn on Peltier cooler for the CCD from Andor Solis software, running on the computer.
8. Arm the laser control box. Care must be taken in setting the laser power - too much power risks damaging the sample.
9. Turn on the digitiser and verify that it is recording data. A few trigger pulses from the microcontroller may be fired to check.

Now, the magneto-PL experiment has been prepped. At this point, the room housing the magnet should be evacuated - closed chambers with a large amount of liquid nitrogen creates a risk of asphyxiation.

Now, to collect data:

1. Charge the capacitor bank using the Variac. Monitor the LED display which reads bank potential and turn the variac off when the target voltage is reached. This determines the magnitude of the peak field during the pulse.
2. Once the capacitor bank is charged, Andor Solis software and the digitiser software are set to record.
3. The routine on the Arduino (i.e. the Arduino sketch) can be modified to set delays and number of datapoints.
4. The routine is uploaded to the Arduino. This triggers a reset on the microcontroller, and the routine starts to execute.

The measurement event lasts 10s of ms, and afterwards the data collected by the acquisition devices need to be saved and labelled.

4.2 Experimental Data

The experiment was carried out at a sample temperature of $5K$. The data was collected over a range of capacitor bank charges from 5 - 70 V in 5V steps.

Data and analysis can be accessed here (Public Github repository).

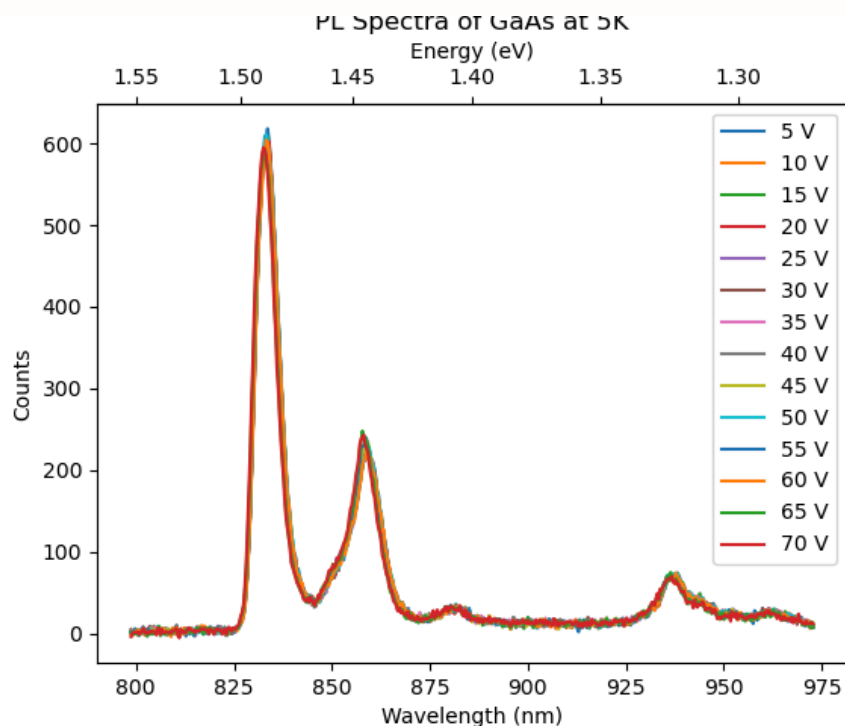


Figure 18: PL Spectra of GaAs at 5K. This plot represents the total dataset collected. It can be observed that apart from the peak near 1.5 eV, the peaks do not shift noticeably with increasing field. Source: Author

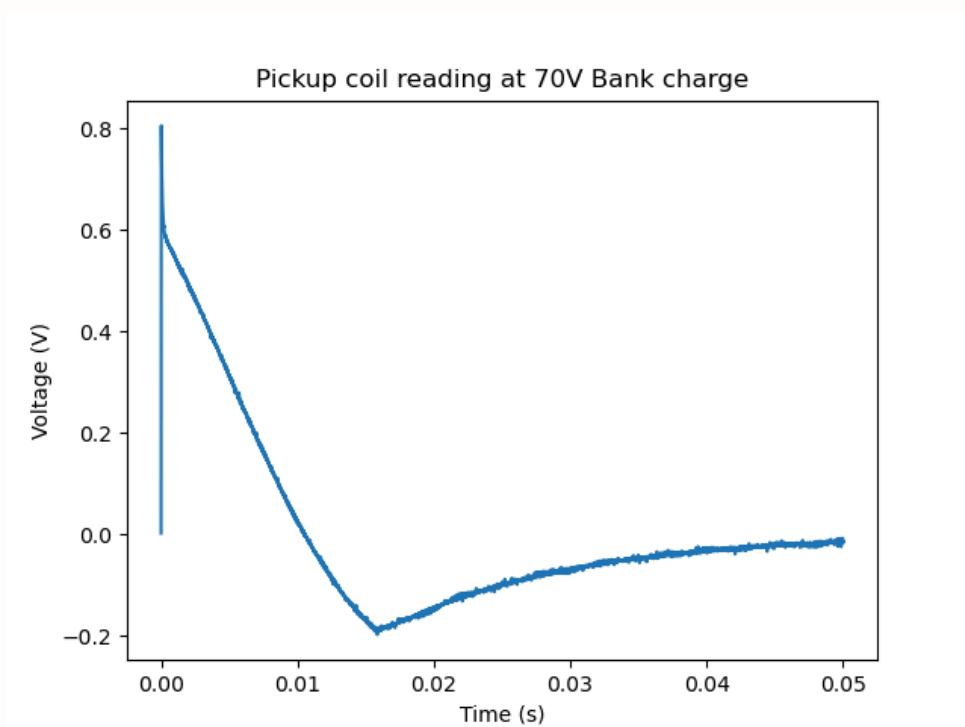


Figure 19: Reading from the pickup coil recorded by the digitiser, 70V bank charge run. Note the recorded negative EMF after the peak. Source: Author

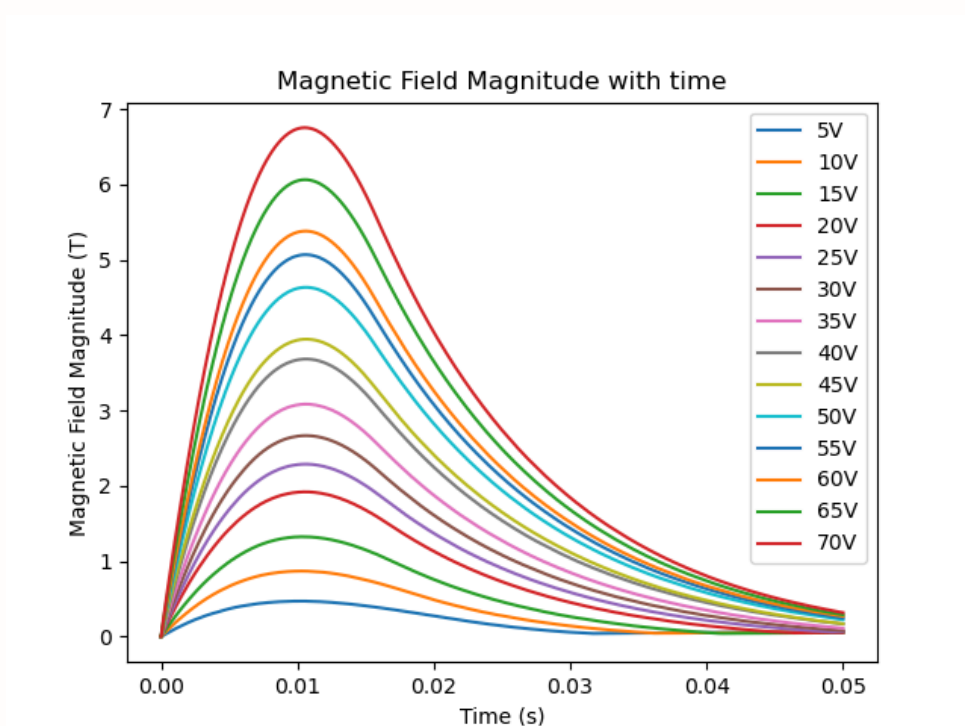


Figure 20: Magnetic Field temporal profiles, integrated from pickup coil data using trapezoidal integration. Source: Author

The peak magnetic fields range from $0.47T$ at 5V to $6.7T$ at 70V.

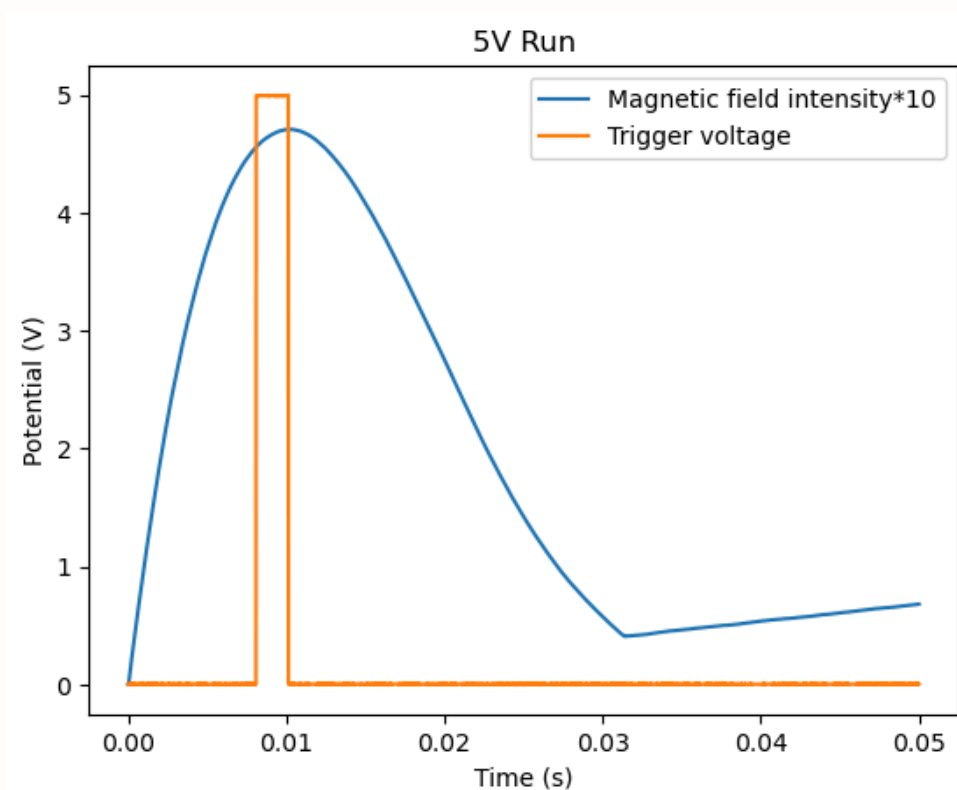


Figure 21: Magnetic Field Profile overlaid on top of camera/laser trigger pulse from microcontroller. The target is to take a spectral snapshot at the peak. Source: Author

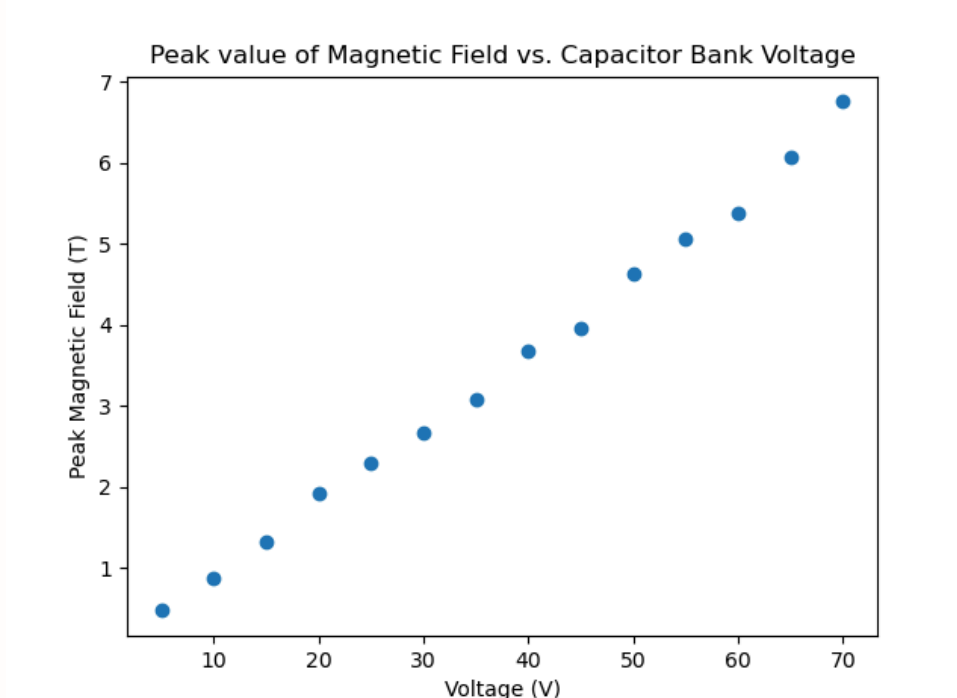


Figure 22: Peak value of magnetic field vs. capacitor bank voltage. Source: Author

Of course, with increasing peak field strength the, the average time rate of change of magnetic field strength over the TTL trigger pulse also increases. It goes from $7.4T s^{-1}$ at 5V to $46.5T s^{-1}$ at 70V. This is negligible when compared to the

timescales over which the light-matter interactions occur - so it is assumed that the field is approximately static when a datapoint is collected at the peak.

4.3 Analysis of Experimental Data

4.3.1 Understanding the spectrum

Note that the PL spectra have three prominent peaks:

1. A peak near 1.5eV . It has a relatively narrow linewidth ($\approx 10\text{meV}$), and a high peak-to-background ratio. We will refer to this as **Peak 1**.
2. A peak near 1.45eV . It has a broader linewidth, and a lower peak-to-background ratio. We will refer to this as **Peak 2**.
3. A peak near 1.30eV . It has a very broad linewidth, and a very low peak-to-background ratio. We will refer to this as **Peak 3**.

A 4th peak can also be discerned between the 1.45eV and 1.30eV peaks. This peak has an extremely poor peak-to-background ratio, and can only barely be made out. We will refer to this as **Peak 4**.

Here, the Full-Width-at-Half-Maximum is a rough measure of the linewidth, but a more careful analysis demands that a Lorentzian be fit to a peak to determine the linewidth accurately.

4.3.2 Fan diagrams

These are plots of the peak energies versus the magnetic field strength.

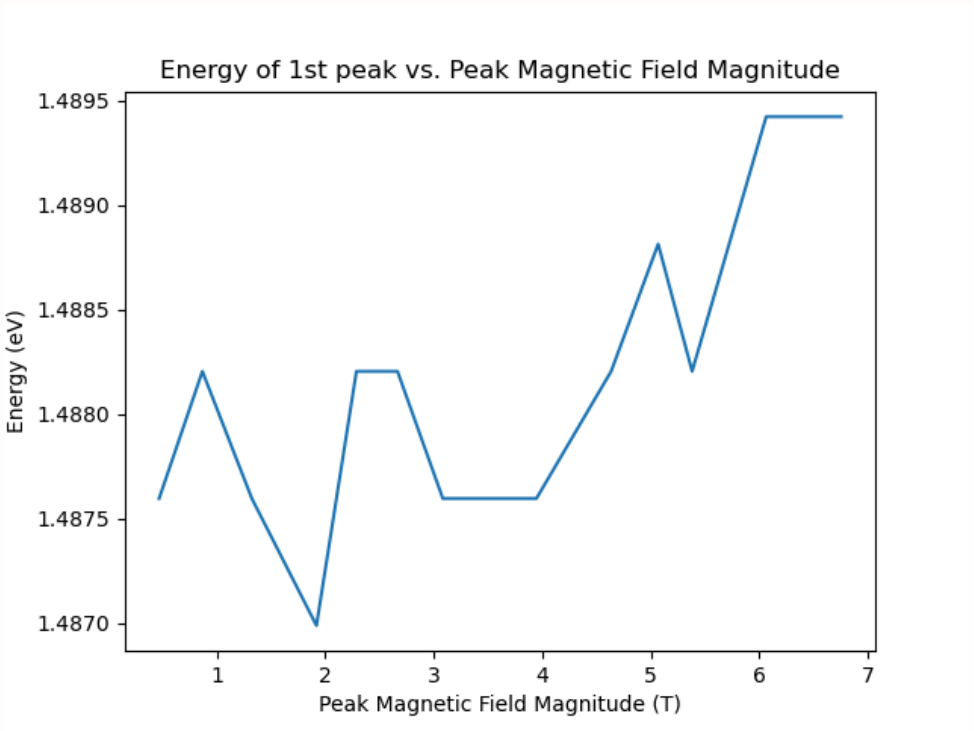


Figure 23: Energy of 1st peak vs. Peak Magnetic Field Magnitude. Source: Author

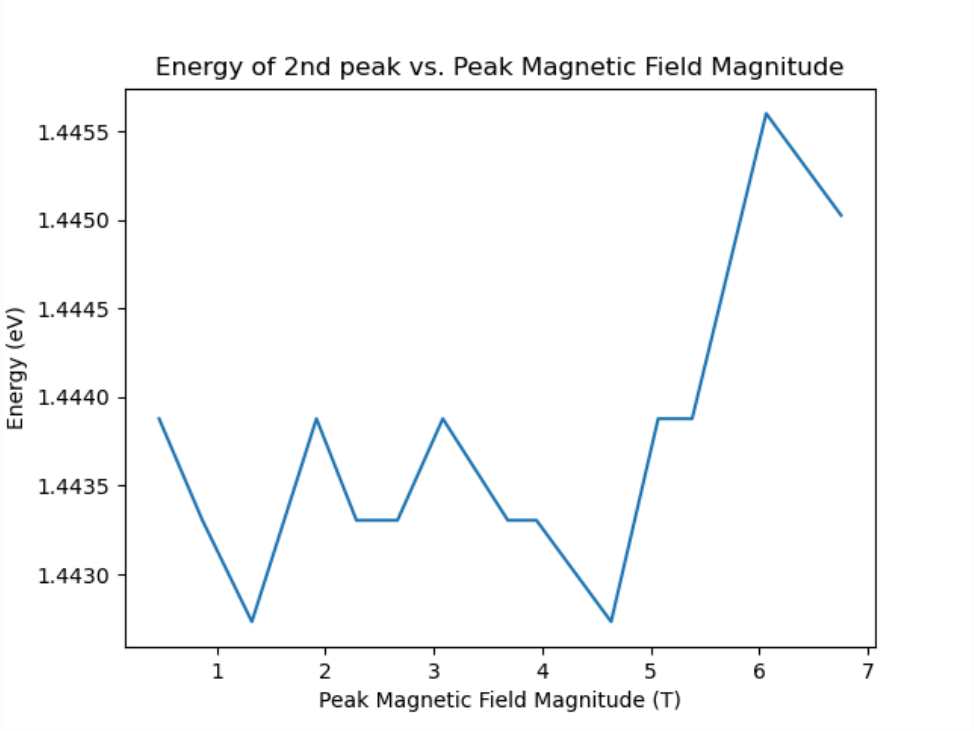


Figure 24: Energy of 2nd peak vs. Peak Magnetic Field Magnitude. Source: Author

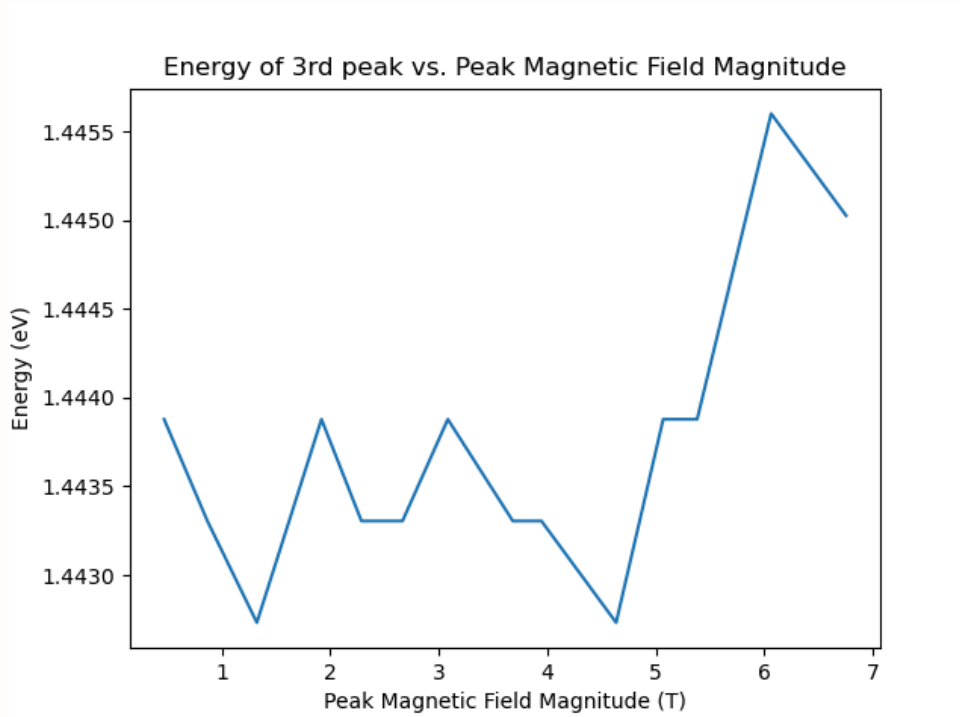


Figure 25: Energy of 3rd peak vs. Peak Magnetic Field Magnitude. Source: Author

It is apparent from this data that we require more dense datapoints, and a larger range in magnetic field strength to have good-quality fits to the data. Since this data was generated to familiarise ourselves with the procedures, this is to be expected. Nevertheless, we will continue with our analysis of this data.

4.3.2.1 Intensity versus Magnetic Field strength

Since this data is available to us as number of counts in the CCD sensor, we are including it for completion. Analysis of this data has not been carried out.

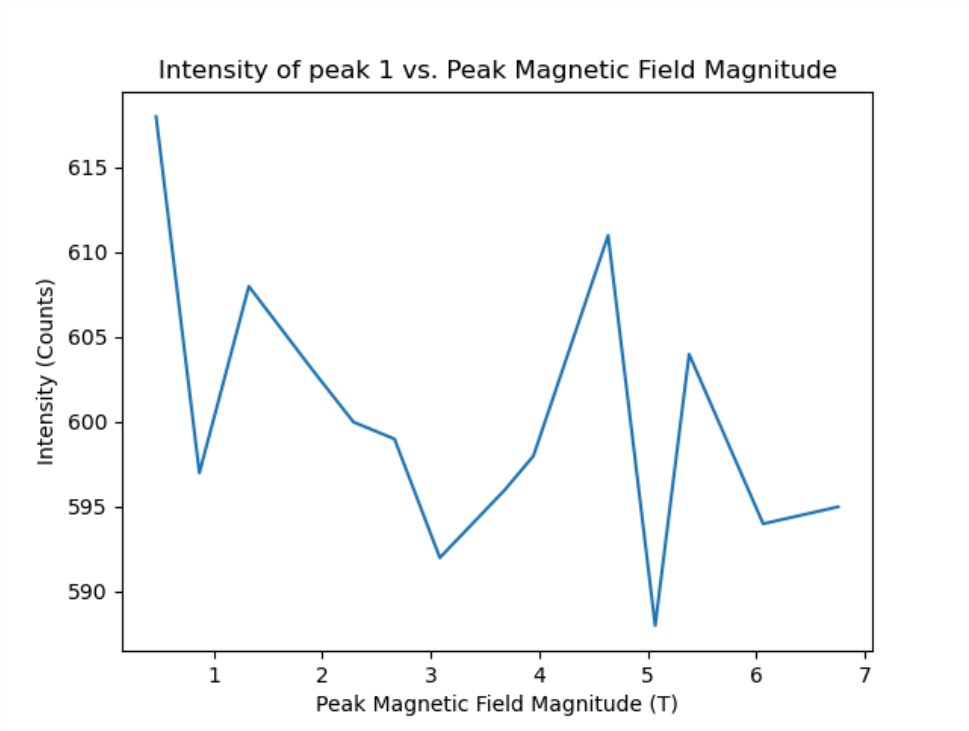


Figure 26: Intensity of peak 1 vs. Peak Magnetic Field Magnitude. Source: Author

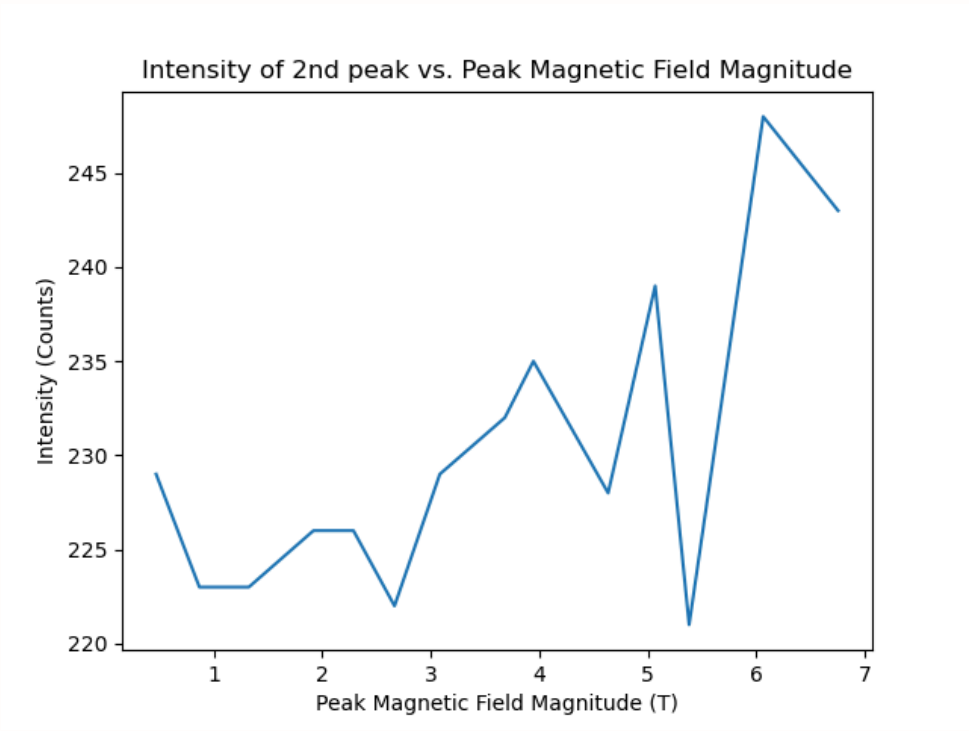


Figure 27: Intensity of peak 2 vs. Peak Magnetic Field Magnitude. Source: Author

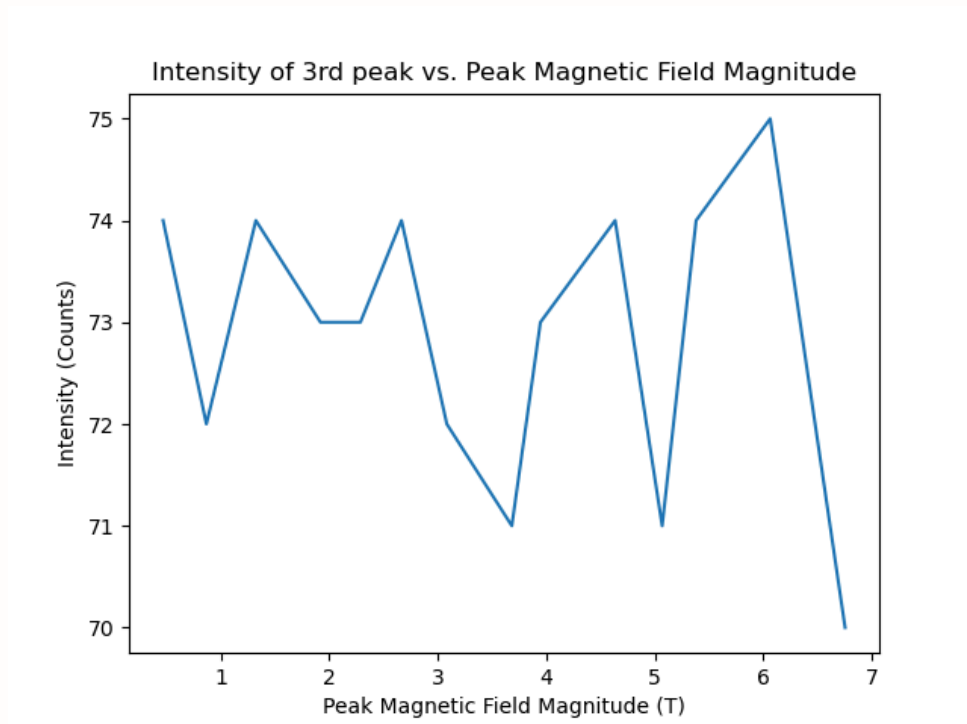


Figure 28: Intensity of peak 3 vs. Peak Magnetic Field Magnitude. Source: Author

4.3.2.2 What are the expected features in the PL spectra?

In high purity, undoped, bulk $GaAs$ at cryogenic temperatures, the broadband 1.4 - 1.5 eV resolves into several lines:

Transition Type	Peak energy $\approx 2K$	Notes
Free exciton ($n=1$)	1.515–1.516 eV	Dominant intrinsic line [29]
Free exciton ($n=2$)	$\approx 1.5195 \text{ eV} - 4.2 \text{ meV}$	Relativity and Spin [30]
Exciton bound to neutral donor (D^0, X)	1.5145 eV	Slightly below free exciton [29]
Exciton bound to ionized donor (D^+, X)	1.5133 eV	[29]
Exciton bound to neutral acceptor (A^0, X)	1.5125 eV	Often carbon-related [29]
Exciton bound to ionized acceptor	1.4886 eV	Deep acceptor complex [29]
Band-to-acceptor ($e-A$)	1.4926 eV	Band-to-impurity recombination [29]
Donor-acceptor pair ($D-A$)	1.4857 eV	Pair recombination band 1 [29]

Table 2: Key low-temperature PL lines in high-purity $GaAs$

Remark: What are these transition types?

While we will not be discussing all these types in detail, a brief summary of some of the important ones are:

- Free Exciton (FE): The radiative decay of the free Wannier exciton. In ultra-pure samples, this may appear as a doublet due to polariton effects (Upper and Lower Polariton Branches), but it is often seen as a single peak.
- Donor-Bound Exciton ($D^0 X$): An exciton trapped at a neutral donor impurity. The localization energy is small, placing this peak slightly below the FE
- Acceptor-Bound Exciton ($A^0 X$): An exciton trapped at a neutral acceptor. Due to the larger mass of the hole and chemical shift, this state is deeper.
- Free-to-Bound Transitions ($e - A^0$): Recombination of a free electron with a hole bound to a neutral acceptor (e.g., Carbon or Zinc), typically found at lower energies.

Clearly, the Free Exciton (FE) peak for $n = 1$ will be the dominant feature.

A zero-field reading was *not taken*, and we will be treating the lowest field (5V charge, 0.47T) as the zero-field reading. It should be close enough to enable us to identify the peaks with the literature value. We discuss them now:

- Peak 1 is clearly the FE ($n = 1$) peak. The (signed) difference from the literature value is: $\Delta_1 = \text{Observed} - \text{Literature} = 1.49\text{eV} - 1.515\text{eV} = -0.025\text{eV}$. It appears that the energies have been lowered by $\approx 0.02\text{eV}$.
- FE ($n = 2$) can not be resolved in our measurements. This may be a resolution issue and should be investigated further.
- Peak 2 may be exciton bound to ionized acceptor peak, or the Band-to-acceptor peak. These are the only peaks that differ from the dominant peak energy by roughly 0.01eV , and thus the only viable choices.
- Peak 3 may be the Donor-acceptor pair peak. This is the only peak that differs from the dominant peak energy by roughly 0.15eV (farthest feature from dominant peak), and thus the only viable choice.

Since we are not studying the effect of impurities in detail, we are concerned with the data of the dominant Peak 1 in this report.

4.3.3 Magnetic regimes traversed

The transition between the low-field (Diamagnetism) and high-field (Landau Quantization) is quantified by the ratio between magnetic confinement energy and coulomb energy. This is given by:

$$\gamma = \frac{\hbar \omega_B}{2R^*} \quad (86)$$

Where $\omega_B = \frac{eB}{\mu}$. The calculated values of γ are plotted.

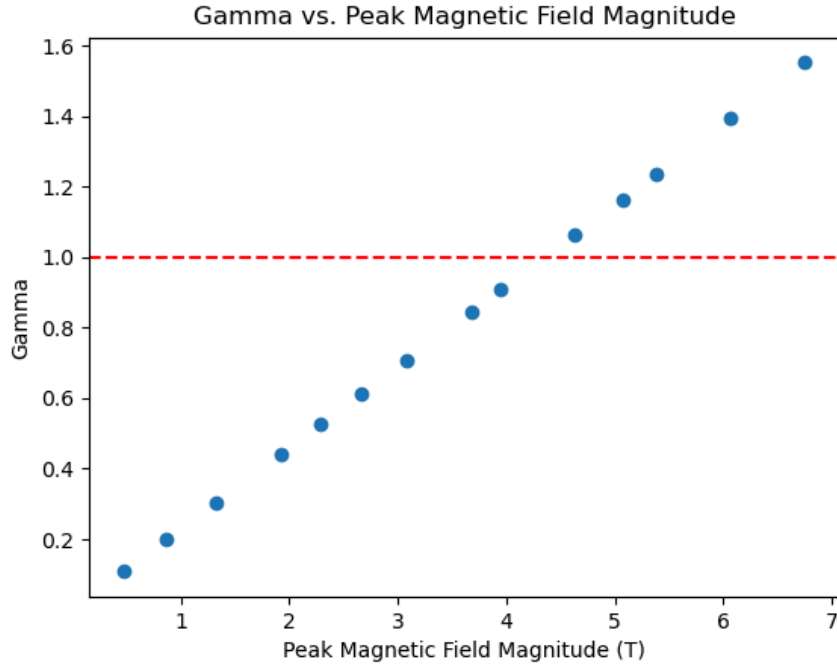


Figure 29: Gamma vs. Peak Magnetic Field Magnitude. Red dashed line demarcates the transition between diamagnetism and quantum confinement. Source: Author

Clearly, the transition between the two regimes occurs between 45V (3.94 T) and 50V (4.63 T). We will be using the average of these two values as our transition point, $B_{tran} \approx 4.285T$.

4.3.4 Determining Diamagnetic shift

Fitting data (peak energy vs. magnetic field) before the transition point to a pure quadratic $f(x) = ax^2 + b$ should yield the diamagnetic shift.

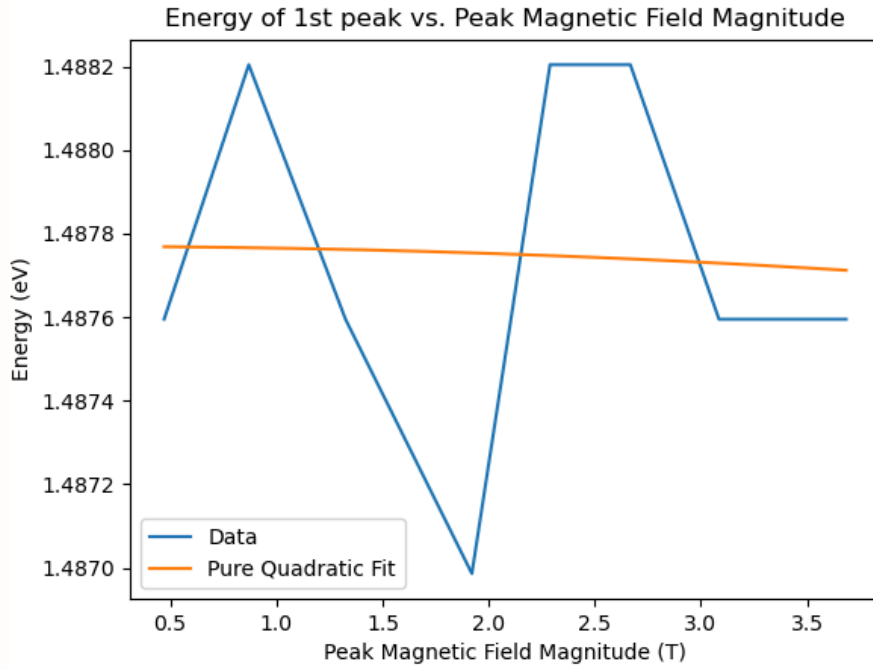


Figure 30: Pure quadratic fit of datapoint with $\gamma < 1$. Source: Author

The fit error in a is $\approx 895.87\%$. Clearly, the fit is non-sensical. More careful data collection is required for a useful fit.

In attempt to derive some information from the data collected, we repeat the analysis by fitting $f(x) = ax^2 + bx + c$ to the entire range of datapoints.

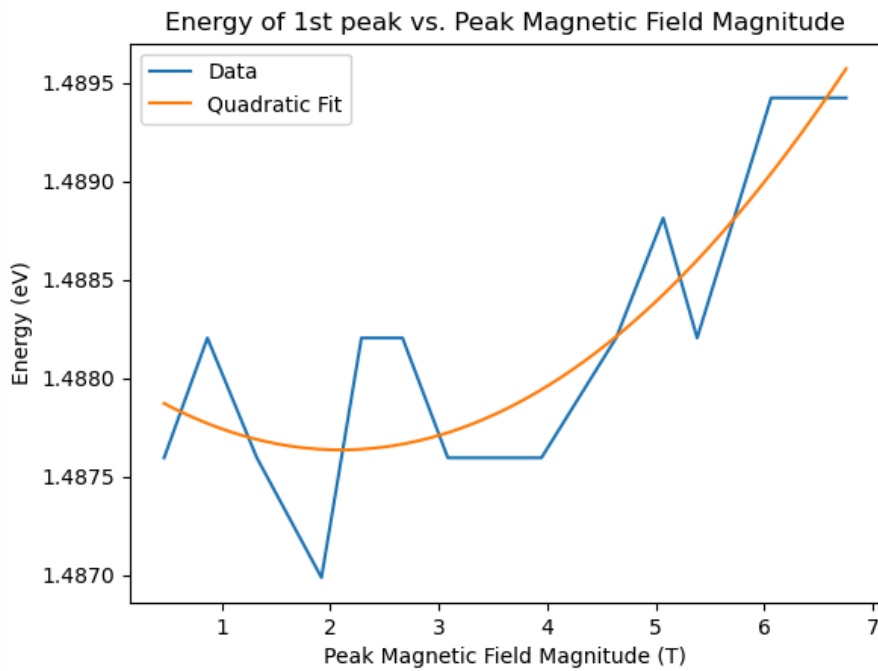


Figure 31: Quadratic fit of data (entire range). Source: Author

Here, the fit coefficients are determined to be:

- $a = 8.92 \times 10^{-5} \pm 37.9\% eV \cdot T^{-2}$
- $b = -3.7 \times 10^{-4} \pm 66.04\% eV \cdot T^{-1}$
- $c = 1.488 \pm 0.025\% eV$

Remark: Unphysical value of b

The diamagnetic shift must *always be positive*. Thus, a negative value of b means a parabola that initially dips - this is not physical. This is probably due to thermal drift in the detector.

The fit is not good quality, but the derived value of diamagnetic shift coefficient can be utilised in the equation:

$$\Delta E_{\{dia\}} = \frac{e^2 B^2}{8\mu} \langle \rho^2 \rangle = \sigma B^2 \quad (87)$$

Where $\langle \rho^2 \rangle$ is the expectation value of the squared coordinate perpendicular to the field direction. For a spherically symmetric 1s state, we have $\langle \rho^2 \rangle = 2(a_B^*)^2$. Thus, this coefficient is directly proportional to the square of the Bohr radius. We find:

$$\sigma = \frac{e^2 a_B^{*2}}{4\mu} \quad (88)$$

Putting in the values, we find a Bohr radius of $11.04 \pm 2.1 nm$. This is in excellent agreement with literature values.

Remark: Alternate analysis

Due to the proximity of the energy of the dominant peak to the literature value of Donor-Acceptor Pair (DAP) or Free-to-Bound (e-A) transition peaks. This could suggest that the sample used is entirely dominated by impurities (carbon-based, as suggested by the literature), and we are not observing FE peaks at all.

4.4 Sources of error

4.4.1 Data collection and Sampling limitations

1. Absence of Zero-Field Calibration: A true zero field ($B = 0T$) reading was not taken during the experiment. The assumption of the 5V charge reading as the zero-field baseline introduces a systematic offset error in the analysis.
2. Insufficient data density/range: The sparse datapoints and insufficient range of datapoints contribute significantly to the large fit errors encountered. Finer steps in magnetic field intensity is required.

4.4.2 Analytical and Fitting Undertainties

1. High statistical error in fits: Since sets of data were not gathered for each magnetic field intensity, the statistical error in the fit is high. There is significant propagated error in the calculated quantities as a consequence of this.
2. Spectral resolution limits: The experiment has failed to resolve the FE ($n = 2$) peak and several other expected impurity signatures. This is due to excessive linewidth broadening or limitation in spectral resolution of the spectrometer.
3. Peak ambiguity and Background noise: Due to broad linewidths and low peak-to-background ratios, several peaks could not be identified accurately.

4.4.3 Instrumental and Hardware limitations

1. Lack of Polarization resolution: Due to the lack of this ability, the effects of Zeeman splitting on Peak 1 could not be discerned in the linear magnetic regime.
2. Capacitor Bank voltage fluctuations: The charging of the capacitor bank is manual and is prone to variation - better control electronics could allow us to reach target peak magnetic field values more accurately and efficiently.

Remark: Resolution Limit Calculation

We can calculate the expected Diamagnetic shift at $7T$ using literature values. If $\sigma \approx 20 \times 10^{-6} \text{ eV} \cdot T^{-2}$ (typical for GaAs FE peak), then the expected shift is:

$$\Delta E_{dia} \approx 1 \text{ meV} \quad (89)$$

The spectrometer resolution is greater than this, but the linewidth may not be - in this case the diamagnetic shift simply can not be resolved.

4.4.4 Methodological assumption

1. Quasi-Static Field assumption: An inherent flaw of the approach is that the pulsed field is assumed to be static over the measurement. This is of course not true. While the effects are minimal due to the timescales of light-matter interaction and the timescales over which the field changes significantly being a few orders of magnitude apart, this is still a source of error. During the signal integration time, the CCD captures pulses from a range of magnetic values, and thus has a “smear”. This broadens the spectral lines.

5 Further work

The next important step for the project, apart from an investigation of the properties of Cu_2O excitons and quantum chaos is the development of an experimental setup that enable us to carry out transmission spectroscopy.

5.1 Fundamental Principles

Transmission spectroscopy is an absorption-based technique that probes the imaginary part of the dielectric function, providing a direct measurement of the oscillator strength of optical transitions. Unlike emission-based techniques, which rely on the relaxation and population statistics of excited states, transmission measures the intrinsic probability of photon-matter interaction.

5.1.1 The Beer-Lambert Law and Absorption Coefficient

In a typical experiment, a monochromatic light beam of intensity $I_0(\omega)$ is incident on a sample of thickness d . The transmitted intensity $I_t(\omega)$ is governed by the Beer-Lambert law:

$$I_t(\omega) = I_0(\omega)(1 - R)^2 e^{-\alpha(\omega)d} \quad (90)$$

where R is the reflectivity (assumed small or accounted for) and $\alpha(\omega)$ is the absorption coefficient. The primary experimental observable is the **optical density (OD)** or absorbance A , defined as:

$$A(\omega) = -\ln\left(\frac{I_t}{I_0}\right) \approx \alpha(\omega)d \quad (91)$$

(ignoring reflection losses for simplicity).

Physically, $\alpha(\omega)$ is related to the imaginary part of the complex dielectric function $\tilde{\epsilon}(\omega) = \epsilon_1 + i\epsilon_2$ and the extinction coefficient κ by:

$$\alpha(\omega) = \frac{2\omega}{c}\kappa(\omega) = \frac{\omega}{nc}\epsilon_2(\omega) \quad (92)$$

where n is the refractive index.

5.2 Theoretical Framework: The Elliott Formula

For a semiconductor hosting Wannier-Mott excitons, the absorption spectrum is described by the **Elliott formula**. The absorption coefficient near the band edge is determined by the overlap of the electron and hole wavefunctions.

The transitions in Cu₂O are unique because the valence band (Γ_7^+) and conduction band (Γ_6^+) both possess even parity. Consequently, the direct dipole transition at $k = 0$ is parity-forbidden ($\langle \Gamma_6^+ | \vec{p} | \Gamma_7^+ \rangle = 0$). Optical transitions become allowed only via:

1. **Quadrupole transitions:** (Weak, relevant for the $1S$ state).
2. **Odd-parity envelope functions:** (Dipole-allowed for P -states).

For the dominant “Yellow” Rydberg series (nP states), the absorption coefficient follows the “forbidden” class of the Elliott formula, where transition probability scales with the derivative of the envelope wavefunction $\nabla \Phi_n(\vec{r})|_{\vec{r}=0}$. The absorption coefficient is given by:

$$\alpha(\hbar \omega) \sim \sum_{n=2}^{\infty} f_n \cdot \mathcal{L}(\hbar \omega - E_n, \Gamma_n) + \alpha_{cont}(\hbar \omega) \quad (93)$$

Where:

- E_n are the exciton eigenenergies given by the Rydberg formula:

$$E_n = E_g - \frac{R_y^*}{(n - \delta)^2} \quad (94)$$

(R_y^* is the excitonic Rydberg constant, approx. 90-100 meV for Cu₂O, and δ is the quantum defect).

- $\mathcal{L}$ is a line-shape function (typically Lorentzian for homogeneous broadening) with linewidth Γ_n .
- f_n is the **oscillator strength**.

Crucially, for these P-type excitons in Cu₂O, the oscillator strength scales as:

$$f_n \sim \frac{n^2 - 1}{n^5} \approx n^{-3} \quad (\text{for large } n) \quad (95)$$

This n^{-3} scaling is a fundamental prediction of the hydrogenic model which transmission spectroscopy can verify directly.

5.3 Advantages of Transmission over Photoluminescence (PL)

While Photoluminescence (PL) is excellent for identifying the lowest-energy relaxed states, Transmission Spectroscopy offers distinct advantages for characterizing the physics of Cu₂O excitons.

5.3.1 Direct Quantification of Oscillator Strength

PL is qualitative; Transmission is quantitative. The intensity of a PL peak, I_{PL} , depends on the population of the excited state $N(E)$ and the radiative decay rate γ_{rad} :

$$I_{PL}(E) \sim N(E) \cdot \gamma_{rad} \quad (96)$$

The population $N(E)$ is heavily influenced by non-radiative relaxation pathways, pumping power, and thermalization (Boltzmann statistics). A state with a giant oscillator strength may appear weak in PL if the excitons relax to a lower energy state before recombining.

In contrast, transmission measures the integrated absorption coefficient, which is directly proportional to the oscillator strength f_n :

$$f_n \sim \int \alpha(E) dE \quad (97)$$

This allows for the direct experimental verification of the n^{-3} scaling law and the extraction of the dipole matrix elements without the ambiguity of carrier dynamics.

5.3.2 Visibility of High-Principal Quantum Number (n) States

PL favors the ground state; Transmission reveals the Rydberg series. In Cu_2O , excitons rapidly thermalize to the lowest available energy state (typically the $1S$ ortho- or para-exciton) via phonon emission. Consequently, the PL spectrum is dominated by $n = 1$ emission and its phonon replicas. High- n Rydberg states (e.g., $n = 10, 20, \dots$) have very low steady-state populations and are often invisible in PL.

In transmission spectroscopy, the ground state population is irrelevant. The technique probes the density of available states from the vacuum. Therefore, transmission allows for the observation of the spectacular Rydberg series up to $n \approx 25$ (or higher in high-quality natural crystals), probing the “Rydberg Blockade” regime where the excitonic radius scales as n^2 and approaches $\sim 1\mu m$.

5.3.3 Analysis of Intrinsic Linewidths

Access to the Scattering Rate (Γ). The linewidth measured in transmission (Γ_{abs}) corresponds directly to the total dephasing rate of the exciton coherence:

$$\Gamma_{abs} = \Gamma_{rad} + \Gamma_{scat} + \Gamma_{inhom} \quad (98)$$

This allows for the study of scattering mechanisms (phonon scattering, impurity scattering) affecting the exciton. In PL, linewidths can be obscured by thermal broadening or spectral diffusion, and often only the phonon-assisted emission lines (which are naturally broad) are strong enough to analyze.

5.3.4 Bulk vs. Surface Sensitivity

PL is often surface-sensitive, especially if the excitation laser has a shallow penetration depth or if surface recombination velocity is high. Transmission acts as a bulk probe, integrating the signal over the entire crystal thickness d . This ensures that the measured Rydberg series properties are representative of the intrinsic crystal lattice rather than surface defects or band-bending effects.

6 Conclusion

In conclusion, I would like to thank Prof. Bhavtosh Bansal for giving me the opportunity to work with him in the Pulsed Magnet Lab for this course. I would also like to thank my seniors in the lab, Siddharth Rout, Aniket Bajaj, and Deepesh Kalauni for their guidance in showing me the ropes of the lab and for their help in the experiments.

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